Wider norms and anchored equality for ZFC consistency

Round 20 research report and reproducible artifacts

Prepared for Vladimir Reshetnikov and the ProveIt repository

25 October 2026

Abstract

A wider common norm for factorial and digit errors gives an explicit quartic in 238 unrestricted integers with 175 quadratic residuals, 5236 collected monomials and coefficient height 3968. The proof recovers signed and unit radix cases before positivity, using a factorial fixed point valid at radius three quarters. A separate factorial-free companion uses the power t^t and has 239 variables, 5137 terms and the same height. For coefficient-height-one equations, replacing equality dummies by products with already anchored unit aliases and contracting internal flow edges give smaller quartics with explicit inverse witness maps, including a 505-variable, 20491-term point and a 503-variable alternative. Whole-expression scalar centers give eleven-integer equations at exact degree and circuit cost 298/147 and 300/145, with separate sharp fixed-frame degree bounds. Complete new ordinary proofs, literal finite arrays, fixed coefficient payloads and independent reproducible checks are supplied. The consistency interpretation composes with identified inherited calibrations; no consistency proof, global optimum or new end-to-end proof-assistant endpoint is claimed.

Contents

1 Results and interpretation **3**

2 Logical scope and literal polynomial definitions **3**
2.1 What the consistency interpretation asserts . . . 3
2.2 The modular structure of the proof . . . 4
2.3 Explicit finite arrays define the direct equations . . . 4
2.4 Arithmetic graphs define the scalar equations . . . 5
2.5 What each measure counts . . . 5

3 A wider common norm and smaller signed quartics **5**
3.1 The retained arithmetic interface . . . 6
3.2 Pell orientation before arithmetic signs . . . 7
3.3 A power lemma valid at signed bases . . . 8
3.4 Nonzero denominators before any sign assumption . . . 9
3.5 Exact factorial digits at a signed radix . . . 9
3.6 A factorial fixed point with radius three quarters . . . 10
3.7 All remaining digits and the refutation . . . 11
3.8 Constructive completeness for the wider sphere . . . 11
3.9 Exact algebra after the arithmetic change . . . 12
3.10 A smaller power base in the factorial-free alternative . . . 12
3.11 Literal selected equations and finite counts . . . 14

4 Height one quartics with anchored equality **14**
4.1 Certified normalization and row orientations . . . 14
4.2 Direct edges for cancelling occurrences . . . 15
4.3 The coefficient-height-one energy . . . 16
4.4 Forward map, inverse map, and uniqueness . . . 16
4.5 Square polarization removes equality dummies . . . 17
4.6 Absorbing equality dummies into anchored products . . . 18
4.7 Contracting an internal flow edge . . . 19
4.8 Selected literal outputs and matched sources . . . 20

5 Polynomial centers and eleven integer equations **21**
5.1 Complete polynomial definitions . . . 21
5.2 An even weight and integral center lemma for the old frame . . . 22
5.3 The complete inherited calibration chain . . . 23
5.4 The incoming rebased frame and two shorter centers . . . 24
5.5 The rebased parent keeps its guard and its late scale proof . . . 25
5.6 Exact degrees at the authentic coefficients . . . 26
5.7 Two exact degree floors in two distinct fixed frames . . . 27
5.8 Measured scalar results and the comparison boundary . . . 28

6 Comparison with Round 19 and the latest inputs **28**
6.1 The immutable comparison snapshot . . . 28
6.2 Direct quartics on unrestricted integers . . . 28
6.3 Height one must be compared on the same source . . . 29
6.4 Other domains and higher degrees . . . 30
6.5 Small variable circuits and fixed coefficient architectures . . . 30
6.6 The limits of the comparison . . . 30

7 Reproduction and evidence boundaries **30**
7.1 Portable entry points and complete definitions . . . 30
7.2 Independent finite checks . . . 31
7.3 The retained logical boundary . . . 32

8 Further questions **32**
8.1 Can the common norm be weakened again . . . 32
8.2 Can contraction and alias choices be optimized together . . . 32
8.3 Can factorial-free arithmetic improve the same frontier . . . 33
8.4 What changes can lower the small scalar degrees . . . 33
8.5 Transporting the higher degree projections . . . 33
8.6 A kernel checked bridge from the literal arrays . . . 33

A Every residual of the balanced principal equation **34**

1 Results and interpretation

The principal equation is a fixed quartic in 238 unrestricted integer variables, with 175 quadratic residuals, 5236 collected monomials and coefficient height 3968. It removes six coordinates and one residual from Round 19's balanced equation, at the same height and with 24 more monomials. Against the newest Shared factorial error balanced equation, it improves all four varying measures: variables, rows, support and height. The 237-variable companion has a denser expansion; a separate 239-variable factorial-free construction is sparser at the same height.

Signed quartic	Variables	Rows	Terms	Height
Balanced principal	238	175	5236	3968
Minimum selected dimension	237	174	18569	7680
Power-base companion	239	175	5137	3968

All three have exact degree four over $\mathbb{Z}$. They are different tradeoffs, not a single equation simultaneously optimal in every measure. The main polynomial is fully defined by its printed residuals in Appendix A and its complete collected coefficient file. It contains no factorial, variable exponent, binomial operation, unspecified coefficient or external predicate.

The main arithmetic change replaces the shared bound $M^2 = 1 + 4(S^2 + \sum R_i^2) + G$ by $M^2 = 2(S^2 + \sum R_i^2) + G$. Nonzeroness is recovered from the signed power request before division; it is no longer a consequence of the norm alone. A radius-three-quarters factorial lemma and the true remainder margin $1/32$ supply the required uniqueness. The separate power-base lemma uses equality of factor counts and positivity of all coefficients to recover bounded label products at $F = t^t$. Neither argument simply assumes the signs or authenticity it is meant to prove.

For height-one quartics, a fresh equality dummy d can be replaced by au, where u is an existing unit alias forced to one. The penalty $C(a, b, au)$ is still nonnegative and has exactly the needed equality zero set. Internal flow-edge contraction supplies another explicit polynomial restoration. Every selected full polynomial is independently collected to verify height one; the zero-set proof alone does not imply that coefficient bound. The selected variable/support pairs are

$$(503, 24193), \quad (505, 20491), \quad (512, 18255),$$
$$(557, 11999), \quad (617, 8318), \quad (713, 7176).$$

All five Round 19 selected points have a successor improving both dimension and support, or dimension with unchanged support. The 503-variable point is a separate smaller-dimension, larger-support tradeoff.

The new eleven-integer scalar points are exact degree/cost 298/147 and 300/145. The first improves both degree and cost over the last 300/148 delivery, but the newest incoming source already has degree 298 at 150 gates. The new achievement against that source is three fewer gates. A second restricted degree theorem identifies 298 as the minimum under constant radical weights and arbitrary integer polynomial centers in that rebased frame; the old frame's distinct minimum remains 300. Neither is a global lower bound on eleven-variable equations.

For each selected endpoint on its stated domain, the ordinary witness proofs compose with the identified calibration to finite syntactic ZFC refutation. Negating the existence equivalence gives unsolvability provably equivalent in ZFC to Con(ZFC). This does not prove consistency or determine whether any of the equations has a zero.

2 Logical scope and literal polynomial definitions

2.1 What the consistency interpretation asserts

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}$ denotes the unrestricted integers. The domain is part of an equation, not a presentation detail. An equation over $\mathbb{N}$ is not thereby an equation over $\mathbb{Z}$, and an unrestricted construction requires its own sign argument or an explicit domain conversion.

Let $\mathrm{Ref}_{\mathrm{ZFC}}(r)$ be the fixed arithmetized predicate saying that r codes a finite syntactic ZFC refutation. For each selected endpoint P on the stated domain D, the claim is

$$\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0 \quad \Longleftrightarrow \quad \exists r \in \mathbb{N} \; \mathrm{Ref}_{\mathrm{ZFC}}(r). \tag{1}$$

The ordinary transformations proved here are composed with the identified inherited calibration. On that proof chain, formalization in ZFC yields

$$\mathrm{ZFC} \vdash [\forall \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) \neq 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{2}$$

This report does not assert that either sentence in (2) is true. In particular it does not prove consistency, decide solvability, establish a global minimum, or supply a new end-to-end proof-assistant kernel check or encoded object-language ZFC derivation.

For the direct families, the retained ordinary logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The first relates the fixed numerical calculus to a finite ZF refutation; the second is the retained ZF-to-ZFC refutation equivalence. The arithmetic front implements that calculus using finite syntax constructors, reciprocal CRT carriers, factorial and power certificates, and range tests. The scalar family instead retains the separately identified source34 native/compiler and refutation calibration, followed by its signed-rank and interpolation reductions. These are two calibrated routes, not interchangeable coefficient tuples.

2.2 The modular structure of the proof

The proof chain has four logically different layers.

1. The inherited calibration identifies a fixed finite arithmetic relation with syntactic refutation. Its provenance and exact numerical payloads are part of the source closure.

2. Unbounded arithmetic arguments prove both witness directions for the certificate relations. These include Pell classification, CRT choices, factorial fixed points, integer signs, and rank or interpolation lemmas.

3. Exact polynomial transformations replace one finite presentation by another. Integral substitutions, restored monic coordinates, graph identities and coefficient collection establish the claimed zero-set correspondence.

4. Finite independent checkers verify the actual emitted arrays and graphs, not merely an abstract instance of a theorem.

A hash certifies file identity, not a theorem; an exhaustive coefficient check certifies an identity, not an unbounded Pell assertion. Each layer is used only for what it establishes. The new sections give complete arguments for this round's transformations; the retained source closure supplies the inherited arithmetic and logical inputs at their stated boundary.

2.3 Explicit finite arrays define the direct equations

For a direct endpoint, the ordered `variables` array in `equation.json` fixes the independent coordinates $x_0, \ldots, x_{n-1}$. A term encoded as $[c, [i_1, \ldots, i_d]]$ denotes $c \prod_{j=1}^{d} x_{i_j}$; the empty index list denotes the constant c. Repeated indices are retained and give ordinary powers. The m residual rows $R_0, \ldots, R_{m-1}$ have integer coefficients and degree at most two. The equation is exactly

$$P(\boldsymbol{x}) = \sum_{j=0}^{m-1} R_j(\boldsymbol{x})^2 = 0. \tag{3}$$

Every coefficient and every coordinate appears in a finite supplied array. The adjacent `expanded_polynomial.json` stores the fully collected polynomial, including its constant term. There is no factorial, variable exponent, division, binomial coefficient, bounded quantifier, or sequence symbol among its algebraic primitives. Such expressions occur only in the proof explaining its zeros.

For either domain $D \subseteq \mathbb{Z}$, equation (3) is equivalent to $R_j(\boldsymbol{x}) = 0$ for every j, because each square is nonnegative. Exact degree four is established by a collected nonzero quartic coefficient, not merely by the upper bound $\deg R_j \leq 2$. All independent coordinates are counted; named arithmetic abbreviations are not new unknowns.

The height-one endpoints are defined directly by their collected coefficient arrays. Their graph certificates prove the displayed energy identity and its zero-set correspondence. The independent checker reconstructs the energy and compares every collected coefficient. The height claim therefore concerns the final polynomial, not only its generating rows.

2.4 Arithmetic graphs define the scalar equations

The scalar graph leaves are eleven independent unrestricted integer variables, ordinary integer constants, or six fixed authentic numerical ports. Each internal node is binary addition, subtraction, or multiplication. An acyclic graph with one designated output therefore denotes a unique element of an ordinary polynomial ring over $\mathbb{Z}$. Registers abbreviate polynomials; they do not introduce hidden existential variables. There is no division or radical gate.

The source34 tuple is the inherited `mixed-parity-unrounded-1064921` tuple at coefficient source commit

`2605669a532342f153b13eb6b4a0af4b54f5fa37`.

Its ports include $\beta = 2^{1064921} - 1$, cone_A $= 2^{1113}$, cone_B $= 2^{666104}$, and the complete literal K_0, M_C, M_F payloads. The retained M_F already includes its native-value-plus-β shift. It must not be shifted twice or replaced by a port from a positive-domain tuple. The payload files contain complete minimal unsigned little-endian integers and are bound by byte length, bit length and SHA-256; neither a hash nor an ellipsis is substituted for a coefficient. The finite inherited compiler recipe reconstructs those values.

2.5 What each measure counts

For a nonzero collected polynomial $P = \sum_\alpha c_\alpha x^\alpha$, write

$$\deg P = \max_{c_\alpha \neq 0} |\alpha|, \qquad |\operatorname{supp} P| = \#\{\alpha : c_\alpha \neq 0\}, \qquad \operatorname{ht}(P) = \max_\alpha |c_\alpha|.$$

Support includes a nonzero constant coefficient. The number of residual rows is distinct from support and from the number of operations. Scalar cost counts live binary graph gates, including squares and multiplication by fixed constants; copying a register and reading a leaf are free. Dead nodes do not count. Unless an expanded scalar polynomial is materialized, its support and total coefficient height are unmeasured, not zero.

A point dominates another only when it is no worse in every displayed measure and strictly better in at least one, with domain and degree held fixed where required. This is a finite comparison among specified presentations. Height one, smaller dimension, sparse support and low graph cost are different objectives, and no single global simplicity ordering is claimed.

3 A wider common norm and smaller signed quartics

Every coordinate in this section is an unrestricted integer. The immediate arithmetic input is the Shared factorial error construction, including its signed affine radix, full fourteen-pair orientation pool, deterministic Pell comparison center and eleven normalized digit remainders. Its source

is not the positive-radix odd-square chart of Round 19. The new argument weakens its shared norm while keeping the signed bootstrap valid; exact row and coordinate reductions then lower the dimension and coefficient height. The distinction matters because the wider norm no longer proves its denominator nonzero by itself.

The inherited arithmetic interface and the signed-base power proof are reproduced first, with the new norm displayed explicitly. The subsequent subsections prove every additional implication needed for the change, including the unit-radix and zero-factorial branches. The retained logical calibration is exactly the C222/ZF/ZFC chain described in Section 2.

3.1 The retained arithmetic interface

Write $t = 16 + d^2$, where d is an unrestricted integer. The principal arithmetic quantities are

$$F, \quad B, \quad V, \quad x, \quad y, \quad N, \quad q, \quad M, \qquad H = 1 + \sum_{\nu=1}^{9} a_\nu.$$

The a_ν are the nine actual CRT representatives. In the stored source, several are a coordinate plus one; it is their actual values, rather than their shifted coordinates, that will become nonnegative.

The retained exact-power requests have the intended meanings

$$V = (F^2 + 1)^t, \qquad N = B^x, \qquad q = (BH)^x, \qquad M = q^x, \tag{4}$$

together with the powers used in the binomial numerators and denominators. There are eighteen requests, sharing fourteen independent principal Pell pairs. At this point these displays are requested meanings, not assumptions. Their interpretation is recovered in a specified order below.

The beta spacing is left literally unchanged:

$$y = V(V - 1). \tag{5}$$

The endpoint equations include

$$x = F + 1 + (y + 1)u_0, \qquad x = 1 + \big(1 + (t + 1)y\big)u_t. \tag{6}$$

Initially u_0, u_t, x, F, B, q may have either sign.

The eleven normalized remainder registers fall into four groups:

Denominator	Meaning	Number
q^t	Factorial digit and encoded index	2
q^{t+1}	Four syntax representatives and derived label	5
q^{y+1}	Two pending-multiset representatives	2
q^x	Two beta quotients	2

The existing ratio equations are

$$q^t v_t = M, \qquad q^{t+1} v_{t+1} = M, \qquad q^{y+1} v_y = M. \tag{7}$$

Each displayed power in these equations is a retained output register. The quotient for the last group is the constant one.

For clarity, put

$$G(a, b, c) = a^2 + b^2 + c(c + 1).$$

This form is nonnegative at every integer triple and represents every nonnegative integer. The latter is the familiar consequence of the three-square theorem applied to $4n + 1$: exactly one square is odd, and the two even squares can be divided by four. This is the same guard form already used in the parent.

The inherited signed affine-radix core replaces three separate initial scalar sign certificates by the first three rows below. This round changes only its last row, from $M^2 = 1 + 4(S^2 + \sum R_i^2) + G$ to

$$q = 2z + 1, \tag{8}$$
$$Bv = M, \tag{9}$$
$$MF = vV + S, \tag{10}$$
$$M^2 = 2\left(S^2 + \sum_{i=1}^{11} R_i^2\right) + G(a, b, c). \tag{11}$$

The integers v, S normalize the factorial error at the same denominator M as the other remainders. No inequality, division, factorial, or variable power has been added to the literal polynomial system. Equations (8)–(11) are polynomial rows of degree at most two in their actual registers.

3.2 Pell orientation before arithmetic signs

For a signed integer center A, define the Pell polynomials by

$$X_0(A) = 1, \quad Y_0(A) = 0, \qquad X_{k+1} = AX_k + (A^2 - 1)Y_k, \quad Y_{k+1} = X_k + AY_k.$$

For negative indices use $X_{-k} = X_k$, $Y_{-k} = -Y_k$. The retained frame bounds every actual base, requested index, and proposed power output in absolute value by an integer W, with

$$W \geq 258, \qquad |A| > W^W, \qquad |A - 1| > 4W^3. \tag{12}$$

These are consequences of square-based frame equations. Their proof does not assume signs for F, B, q, x, or the power outputs.

The deterministic comparison center is retained from the supplied predecessor proof. To distinguish its auxiliary pair from the arithmetic quantity H, write it as

$$H_{\mathrm{p}}^2 - (A^2 - 1)J_{\mathrm{p}}^2 = 1, \qquad J_{\mathrm{p}} = C^2(2e + 1),$$

where (U, C) is the principal anchor pair with requested index h. The squared cap says

$$C^2 = h^2 + 1 + \sum_{\text{other pairs}} Y_i^2 + G(a', b', c').$$

It gives $C \neq 0$, $|h| < |C|$, and strict domination of all other principal ordinates. The comparison center is defined by

$$f = (1 - A)H_{\mathrm{p}}, \qquad D = A + H_{\mathrm{p}}f = 1 - (A - 1)(A^2 - 1)J_{\mathrm{p}}^2.$$

Thus $|D| > 1$, $D \equiv A \pmod{H_{\mathrm{p}}}$, and $D \equiv 1 \pmod{C}$. The comparison ordinate residue is $R = h + Ck$.

Here is the part of its proof that matters for compatibility. Classify the anchor and common Pell ordinates at $|A|$ by positive indices i, l. For $i \geq 2$, the squared divisibility condition implies $|C| \mid l$ and $l > i$. The abscissa comparison gives the comparison index $j \equiv \pm i \pmod{2l}$; the new ordinate residue gives $j \equiv \pm h \pmod{|C|}$. Consequently $i \equiv \pm h \pmod{|C|}$. The growth bound

$$|C| = Y_i(|A|) \geq i + 2|A| - 2$$

and $|h| < W$ imply $i + |h| < |C|$, so $i = |h| < W$. All other classified indices are smaller by the cap.

The exceptional case $i = 1$ has $|C| = 1$. The cap then forces $h = 0$ and every other ordinate to vanish. Reducing the retained orientation pool modulo $A - 1$ gives a nonnegative small error sum.

Its anchor ordinate contributes one, so it is strictly positive, while (12) makes it strictly smaller than $|A-1|$. Its asserted divisibility is impossible. This exceptional branch therefore supplies no zero.

All fourteen principal pairs retain both-coordinate orientation: the squared abscissa errors and squared ordinate-index errors remain in the pool. After classification their small residues have errors at most two and strictly less than $2W$, respectively. The resulting nonnegative residue sum is less than $4 \cdot 14(W^2+1) < 4W^3 < |A-1|$. Divisibility forces every error to vanish. Thus each principal pair is exactly $(X_k(A), Y_k(A))$ at its signed requested index.

This use of all fourteen pairs is deliberate. A separate predecessor reduction keeps fewer abscissa tests, but its hypotheses cannot simply be assumed in the signed bootstrap below. The present construction does not make that deletion. The complete retained center and frame proofs are included in the proof closure.

3.3 A power lemma valid at signed bases

Lemma 3.1. *Assume* (12). *Let* $|n|, |k|, |m| < W$, *with* $|n| \geq 2$, *and set*

$$T = 2An - n^2 - 1.$$

If the pair is canonically oriented at k *and its power row asserts*

$$X_k(A) + (n-A)Y_k(A) \equiv m \pmod{T},$$

then $k \geq 0$ *and* $m = n^k$. *For* $n = \epsilon \in \{1, -1\}$, *the same row instead gives* $m = \epsilon^k$ *for every integer* k, *including negative* k.

Proof. For $|n| \geq 2$,

$$|T| > 4W^W - (W-1)^2 - 1 > W^W + W.$$

Modulo T, one has $(n-A)^2 = A^2 - 1$ and $n(2A-n) = 1$. The Pell identities therefore give n^k at nonnegative indices and its multiplicative inverse at negative indices. If $k \geq 0$, the congruence has

$$|m - n^k| < W + W^W < |T|,$$

so it is equality. If $k = -j < 0$, multiplication by n^j gives $mn^j \equiv 1 \pmod{T}$. But

$$|mn^j - 1| \leq (W-1)^W + 1 < |T|,$$

forcing the impossible integer equality $mn^j = 1$.

For a unit base, $T = 2A\epsilon - 2$ and $2A \equiv 2\epsilon \pmod{T}$. The sequence $f_k = X_k(A) + (\epsilon - A)Y_k(A)$ has initial values $f_0 = 1, f_1 = \epsilon$ and satisfies $f_{k+1} = 2Af_k - f_{k-1}$. This recurrence runs backwards as well as forwards. Hence $f_k \equiv \epsilon^k \pmod{T}$ for every signed integer k. Finally $|T| > 2W^W - 2 > W$, whereas $|m - \epsilon^k| \leq W$, giving the stated equality. □

The base -1 case is essential: it cannot be replaced by a lemma that presupposes a nonnegative exponent. The literal request for $M = q^x$ has the row

$$-M + X_x(A) - AY_x(A) + qY_x(A) - T_q\eta = 0.$$

Both M and q, and the actual base BH of the self-radix request, are included in the frame. Thus the lemma applies to the unshifted arithmetic values claimed here.

3.4 Nonzero denominators before any sign assumption

The weakened norm no longer has a constant one on its right-hand side. It therefore does not, by itself, imply $M \neq 0$. The correct order is as follows. The affine oddness row gives $q \neq 0$. Apply Lemma 3.1 immediately to the request for M at base q. For $|q| \geq 2$ it gives $x \geq 0$ and $M = q^x$; for $q = \pm 1$ it gives $M = q^x$ at every signed index. Either way $M \neq 0$. The request at the already positive index t gives $V = (F^2+1)^t \geq 1$, including $F = 0$.

From $Bv = M$ we obtain $Bv \neq 0$. Set $b = |B|$ and $s = BF - V$. The other new row gives $S = vs$, and the sphere gives

$$|s| \leq b/\sqrt{2} < 3b/4. \tag{13}$$

If $F = 0$, then $V = 1$ and $s = -1$, so $b \geq 2$. If $f = |F| \geq 1$, opposite signs of B, F would force $V = BF - s < -b + 3b/4 < 0$. Thus $BF = bf > 0$. Moreover $V < b(f + 3/4)$. For every integer $f \geq 1$,

$$(f^2+1)^t \geq (f^2+1)^2 > (f+1)(f+3/4),$$

since the last difference is $f^4 + f^2 - 7f/4 + 1/4 > 0$. It follows that $b \geq f + 2 \geq 3$, and therefore

$$V < b(f+3/4) \leq b(b-5/4) < b^2 - b < b^2. \tag{14}$$

In particular every branch has $b \geq 2$.

If $q = \pm 1$, its power request gives $|M| = 1$, contradicting $Bv = M$. Hence $Q := |q| \geq 3$, and $|q+1| \geq 2$. Every actual representative a_ν is an exponent requested at base $q+1$, so Lemma 3.1 gives $a_\nu \geq 0$ and $H \geq 1$. The same argument gives $h \geq 0$, while the semantic anchor has already excluded $h = 0$. The request at base q now gives $x \geq 0$ and $M = q^x$. The retained denominator-ratio equation $q^{t+1} v_{t+1} = M$, interpreted as exact nonzero powers and taken in absolute value, forces $x \geq t + 1 \geq 17$. Consequently the self-radix request has base BH of magnitude at least two and gives

$$q = (BH)^x, \qquad Q = (bH)^x \geq b^x \geq b^{t+1}. \tag{15}$$

If $F = 0$, we have so far proved only $b \geq 2$. But q is odd and $x > 0$, so b is odd; hence $b \geq 3$. This parity step is necessary when widening the former radius $1/2$.

3.5 Exact factorial digits at a signed radix

For each normalized row, $L_i v_i = M \neq 0$ and $v_i U_i = MQ_i + R_i$. Hence $v_i \mid R_i$, and $r_i = R_i/v_i$ is an integer satisfying

$$U_i = L_i Q_i + r_i, \qquad |r_i| \leq |L_i|/\sqrt{2} < 3|L_i|/4. \tag{16}$$

This uses no positivity of L_i or r_i.

For the factorial digit, set $n = F^2 + t$ and $C_j = \binom{n}{j}$. Its exact numerator request gives $U = (q+1)^n$, while its restored digit row gives

$$U = q^t(B + q\eta) + r.$$

The signed low polynomial $\rho = \sum_{j=0}^{t-1} C_j q^j$ satisfies

$$|\rho| < Q^t \frac{\max_{j<t} C_j}{Q-1}.$$

For $F \neq 0$, the elementary coefficient estimate $C_j \leq (F^2+1)^t = V$ holds for all $j \leq t$. To prove its full range, induct on t in $\binom{m+t}{j} \leq (m+1)^t$, for $m \geq 1$ and $0 \leq j \leq t$. Pascal's identity handles the next row through $j = t$, since $2 \leq m + 1$. Its new edge is bounded by

$$\binom{m+t+1}{t+1} = \prod_{i=1}^{t+1}(1 + m/i) \leq (m+1)^{t+1}.$$

The initial case $t = 0$ is immediate. Equations (14)–(15) now give

$$|\rho| < Q^t \frac{V}{Q-1} < Q^t \frac{2}{b^{t-1}} < Q^t/32.$$

For $F = 0$ one must not use $V = 1$ as a coefficient bound. Instead $C_j = \binom{t}{j} \le 2^t$ and $b \ge 3$, so

$$|\rho| < Q^t \frac{2^t}{3^{t+1}-1} < Q^t (2/3)^{t+1} < Q^t/32.$$

The difference $r - \rho$ is divisible by the possibly negative integer q^t, but its absolute value is less than $(3/4 + 1/32)Q^t = 25Q^t/32 < Q^t$. Thus $r = \rho$. Subtracting, dividing by q^t and reducing modulo q gives $B \equiv C_t \pmod q$. For $F \neq 0$, $|B - C_t| \le b + C_t < b + b^2 < Q$; for $F = 0$, $C_t = 1$ and $|B - 1| \le b + 1 < Q$. In both cases,

$$B = \binom{F^2 + t}{t} > 0. \tag{17}$$

The branch $F = 0$ would force $B = 1$, contrary to $b \ge 3$. Thus $F \neq 0$, the earlier common-sign argument gives $F > 0$, and (15) finally gives $q > 0$. The negative-radix branch is excluded by the extracted arithmetic, not by an initial sign assumption. All other requests, including $N = B^x$, now have their ordinary positive meanings.

3.6 A factorial fixed point with radius three quarters

Lemma 3.2. *For integers $t \ge 16$ and $F \ge 1$, let*

$$R_t(F) = \frac{(F^2+1)^t}{\binom{F^2+t}{t}}.$$

Then $|F - R_t(F)| < 3/4$ if and only if $F = t!$. At that solution, $0 < F - R_t(F) < 1/32$.

Proof. Write $K = t!$ and

$$R_t(F) = K \prod_{i=1}^{t} \frac{F^2+1}{F^2+i}.$$

Every factor is nondecreasing with F, and

$$R_t(F) \ge R_t(1) = \frac{2^t}{t+1} > 2t^2 + 3/4.$$

The last inequality holds at 16 and propagates by induction: if $A_t = (t+1)(2t^2 + 3/4)$, then $2A_t - A_{t+1} = 2t^3 - 4t^2 - 37t/4 - 4 > 0$ for $t \ge 16$. The distance hypothesis therefore gives $F > 2t^2$, whence $F \ge 2t^2 + 1$. Since $R_t(F) < K$, the same hypothesis implies the integer bound $F \le K$.

Furthermore $F^2/(F+1) = F - 1 + 1/(F+1) > 2t(t-1)$, so $F^2 > 2t(t-1)(F+1)$. For $u_i = (i-1)/(F^2+i)$,

$$\sum_i u_i < \frac{t(t-1)}{2F^2} < \frac{1}{4(F+1)}.$$

Using $\prod_i (1-u_i) \ge 1 - \sum_i u_i$ gives $R_t(F) > K(1 - 1/[4(F+1)])$. If $K \ge F+1$, this is strictly greater than $F + 3/4$, contradicting the distance hypothesis. Thus $F = K$.

Conversely, at $F = K$ the same product estimate gives

$$0 < K - R_t(K) < \frac{t(t-1)}{2K} < \frac{1}{32}.$$

The final inequality holds at 16 and its left side decreases thereafter. This proves both the equivalence and the stronger canonical margin. □

By (17), estimate (13) is exactly $|F - R_t(F)| < 3/4$. Hence $F = t!$ and $0 < s = BF - V < B/32$. The old ceiling witnesses s and $B - s - 1$ are nonnegative integers. Also $q \ge B^{t+1} > B$, so $q - B - 1 \ge 0$. Fresh ternary G witnesses restore all three deleted initial scalar sign certificates.

3.7 All remaining digits and the refutation

Only now use the true positive factorial front in the beta endpoints. Since $y = V(V-1) > F+2$, the first endpoint with $u_0 \le -1$ would give $x \le F - y < 0$. Thus $u_0 \ge 0$ and $x \ge F+1 \ge 2$. The last endpoint then forces $u_t \ge 1$, and

$$x \ge 2 + (t+1)y. \tag{18}$$

All four denominator levels are now positive divisors of $M = q^x$. The strict amplification margin is unchanged. With $\mathcal{R} = 2 + t + x + y + F$, one has $\mathcal{R} < 2x$ and $x \ge 16! + 1 > 2^{32}$; therefore

$$\mathcal{R}^{2^{23}} < (2x)^{2^{23}} < 2^x.$$

For the last inequality, the function $x/\log_2(2x)$ increases for $x \ge 2^{32}$, and at 2^{32} it exceeds 2^{23}. This is precisely the retained local precision margin.

The encoded-index congruence is $B \mid C_0 + F^2 + 1$. Since $B > F^2 + 1$ and $C_0 \ge 0$, we have $C_0 > 0$ and $H \ge 2$. Each ordinary representative $a \ge 0$ has $\binom{a}{j} \le \max(1,a)^k \le H^x$ at its range position $k \le x$, including $a = j = 0$. The retained constructor degree ledger gives $h \le H^{2^{22}}$, while the factorial and endpoint inequalities give $2^{22}(t+1) < x$. Thus all needed derived-label coefficients also are at most H^x. Since $q = B^x H^x$, every genuine remaining low tail obeys

$$0 \le \rho_i < L_i \frac{H^x}{q-1} < L_i \frac{2}{B^x} < L_i/32.$$

Equation (16) and the same divisible-difference argument force $r_i = \rho_i$. Every original interval slack $L_i - r_i - 1$ is nonnegative and admits its old guard witnesses. The quotient congruences are exactly the retained N-divisibility range tests, with $N = B^x \mid q$. The unchanged prime-power extraction, constructor equations and Horn-flow theorem therefore recover a finite C222 refutation, and the named calibration gives a finite ZFC refutation. This recovers a complete calibrated witness, not just some necessary arithmetic conditions. Stored representative-minus-one coordinates may still be -1; only the actual semantic representatives must be nonnegative.

3.8 Constructive completeness for the wider sphere

Start with a finite calibrated refutation. The retained cofinal construction chooses a sufficiently large $t = 16 + d^2$, sets $F = t!$, $V = (F^2+1)^t$, $y = V(V-1)$, and chooses a sufficiently large beta-CRT code x. The true beta quotients are nonnegative and less than x, since their denominators are at least two. All original strict range witnesses therefore exist.

Every factor $1 + F^2/i$ is odd because F is even and $i \mid F$. Thus B, $N = B^x$ and the precision modulus N^2 are odd. Adding N^2 to a representative preserves its CRT residues and reverses parity. Choose the dominant representative a in $[2N^2, 4N^2)$ with parity opposite to b, retaining the others in $(0, 2N^2]$. The literal constructors give $h \ge 2a^2$ and $h \equiv 1 + a + b \pmod 2$. Consequently h is even and remains a dominant semantic anchor. Choose a third independent representative's parity so the sum of all nine is even, making H odd. These choices preserve all precision bounds. Then $q = (BH)^x$ is odd, and $z = (q-1)/2$ is an integer. Affine oddness requires no parity condition on x.

For the deterministic comparison center take an admissible positive A, $C = Y_h(A)$ and $U = X_h(A)$. Since h is even, C is even and U is odd. Set $l = hC$. Pell composition gives $J_{\mathrm{p}} = Y_l(A) = CY_C(U)$. The Pell binomial expansion and $U^2 - 1 = (A^2-1)C^2$ imply

$$\frac{Y_C(U)}{C} \equiv U^{C-1} \pmod C.$$

Indeed the zeroth term after division by C is U^{C-1}; every higher term contains a factor $(U^2-1)^j/C$, hence at least one factor C. Thus J_{p}/C^2 is an odd integer and supplies e. Set $H_{\mathrm{p}} = X_l(A)$,

$f = (1 - A)H_{\mathrm{p}}$ and $D = A + H_{\mathrm{p}}f$. The pair $(X_h(D), Y_h(D))$ supplies both required comparison residues, including when $D < 0$. The retained cofinal frame and cap construction provides its other witnesses. Canonical pairs have residues $(1, k)$ modulo $A - 1$, so the full orientation pool holds.

Finally $B \mid q \mid M$. Set $v = M/B$ and $S = v(BF - V)$. Lemma 3.2 gives $0 < S < M/32$. For each true remainder take $R_i = (M/L_i)r_i$, with $0 \le R_i < M/32$. Thus

$$2\left(S^2 + \sum_{i=1}^{11} R_i^2\right) < \frac{24}{1024}M^2 = \frac{3}{128}M^2 < M^2.$$

The nonnegative integer difference has a G representation, supplying the new sphere row. Every other source witness is retained. This constructs a new zero from every finite refutation; it is not a bijection between arbitrary old and new ternary-guard assignments.

3.9 Exact algebra after the arithmetic change

The remaining reductions use three finite operations. An integral affine coordinate shear has its explicit integral inverse. An elementary row change $R_i \leftarrow R_i + gR_j$, $i \neq j$, preserves simultaneous zeros and is inverted with $-g$. A unit-pivot row $\epsilon x_v + f(\widehat{x_v}) = 0$, $\epsilon = \pm 1$, uniquely restores $x_v = -\epsilon f$ and can be deleted after that substitution in every other row. The certificate verifies that the proposed pivot is independent of x_v, that every transformed row remains quadratic when claimed, and that all coordinate indices are compacted consistently.

These transformations preserve zero existence but need not preserve the value of the sum of squares away from zeros. The final polynomial is the sum of squares of the final literal residuals, not an asserted coefficientwise identity with an earlier quartic. Combining both arithmetic witness directions with the exact finite transformations gives (1) for every selected endpoint; negation gives (2). The full source calibration is retained, not replaced by the finite coefficient checker.

3.10 A smaller power base in the factorial-free alternative

A separate companion starts from the incoming Factorial-free binomial CRT source rather than from the shared-factorial-error source. Its retained prime-power separation theorem uses

$$R \ge \operatorname{lcm}(1, \dots, t)^2, \qquad B = \binom{R + t}{t},$$

to obtain a suitable full prime-power factor for every row, and to assign every prime-power factor of B to a row for reverse CRT completeness. Its product-label recovery originally required a larger evaluation point. The following elementary refinement permits a simpler authentic power request without changing its local row rules.

Lemma 3.3. *Let $t \ge 2$ be an integer and $F = t^t$. Suppose two multisets of labels from $\{1, \dots, t\}$ have equal products of weights $1 + jF$, and at least one multiset has cardinality at most t. Then the multisets agree, including multiplicities.*

Proof. First the cardinalities must agree. If the smaller count is $d \le t$ and the larger count is at least $d + 1$, the larger product is at least $(F + 1)^{d+1}$, whereas the smaller is at most

$$(1 + tF)^d \le t^d(F + 1)^d.$$

This contradicts $F + 1 > t^t \ge t^d$, including $d = 0$. Thus both cardinalities are the same $d \le t$.

Let $A(z), B(z)$ be the corresponding products of d factors $1 + jz$. When $d = 0$, both polynomials are one. Otherwise every coefficient from degree zero through d is positive, and for $0 \le k \le d$ it is at most

$$\binom{d}{k}t^k = \binom{d}{d - k}t^k \le d^{d-k}t^k \le t^d \le F.$$

Consequently every coefficient of $A - B$ has absolute value strictly less than F: both coefficients being subtracted lie in $\{1, \ldots, F\}$. A strict upper bound on either coefficient separately is unnecessary. If $A - B$ were nonzero, divide it by its lowest nonzero power of z. The resulting nonzero constant coefficient has magnitude less than F, so evaluation at F cannot vanish modulo F. This contradicts $A(F) = B(F)$. Hence $A = B$. The distinct labels give distinct roots $-1/j$, and unique factorization over $\mathbb{Q}$ recovers their multiplicities. Labels zero, if allowed by the arithmetic convention, can first be discarded because their weight is one. □

For this companion choose the cofinal even horizon

$$t = 16 + 2h_0^2, \qquad F = t^t, \tag{19}$$

using the already present genuine power request at index t with base t. These are the meanings of existing quadratic power certificates, not variable-exponent gates in the final polynomial. The base is an existing register, so its expression no longer needs the incoming $2(t + 1)$ construction. The literal delta consists of two raw rows: $t_0 = h_0^2$ becomes $t_0 = 2h_0^2$, and the growth base $2t_0 + 34$ becomes $t_0 + 16$. Every horizon use inherits its changed value through the same shared t_0 register. The rebuilt frame includes the actual new base and every other required request atom; the independent source-delta checker verifies that complete propagation.

We check all new arithmetic compatibility conditions. For $t \geq 16$, $F = t^t > t!$, $F \geq 16^{16} > 2^{32}$ and $v_2(F) = t\, v_2(t) \geq t$, because t is even. Taking $R = F^2$ therefore exceeds the retained CRT threshold. For $1 \leq i \leq t$, the inequality $v_2(F^2) > \log_2 t \geq v_2(i)$ gives $v_2(F^2 + i) = v_2(i)$. Hence

$$v_2\left(\binom{F^2 + t}{t}\right) = \sum_{i=1}^{t} v_2(F^2 + i) - v_2(t!) = 0,$$

so B is odd, as required by the source's representative parity choices.

In completeness the retained beta spacing has the form $y = F(2w + 1)^2$. Choose the odd factor divisible by the odd part of $t!$ and arbitrarily large within that odd divisibility class. Since $v_2(t!) < t \leq v_2(F)$, this makes y an arbitrarily large even multiple of $t!$. The original beta CRT, even-code and odd-square radix construction therefore remains available. The new horizon is cofinal, and the retained padding theorem extends any finite proof to an admissible horizon. The original exclusion of the boundary $y = F$, using the zero initial accumulator and the terminal accumulator one, is unchanged. All other quantitative arguments use only the now verified bounds $F > t!$, $F \geq 7$, $F > 2^{32}$ and the retained endpoint inequalities $x \geq F + 1$ and $x \geq 2 + (t + 1)y$.

The power companion also widens its eleven-remainder norm from $M^2 = 1 + 4\sum R_i^2 + G$ to $M^2 = 2\sum R_i^2 + G$. Here the retained odd-square radix gives $q \geq 1$ independently. The authenticated positive $F = t^t$ gives $y \geq F$, and the numerator requests give nonnegative actual representatives and $H \geq 2$. The raw encoded-index row excludes $B = 0$. The self-radix request then forces $x \geq 0$; the last beta endpoint excludes $x = 0$, and the first excludes a negative initial quotient. Hence $x \geq F + 1$ and $x \geq 2 + (t + 1)y$, and $q = |BH|^x \geq 2^x > 1$. Thus $M = q^x \neq 0$ before using the new sphere. The same coefficient estimates as in the retained source, using $F \geq t$, $F \geq 7$ and these endpoint bounds, place every canonical low remainder below $L_i/32$. Dividing each normalized row by its nonzero ratio gives an integral r_i with $|r_i| \leq L_i/\sqrt{2} < 3L_i/4$. The divisible difference from the true remainder has magnitude less than $25L_i/32 < L_i$, so it vanishes. The original bound $|B| < q/2$ and the exact factorial-binomial coefficient bound $C_* < q/32$ still give $B = C_* > 0$ without circularity. For completeness, the canonical remainders give $2\sum R_i^2 < 22M^2/1024 < M^2$, so their nonnegative integer gap has a G representation. This proves both directions of the additional sphere change for this separate source.

For soundness, the unchanged signed bootstrap authenticates the request $F = t^t$, so the retained prime-power theorem applies at $R = F^2$ and Lemma 3.3 supplies the product-label injectivity. The unchanged local calculus and calibration recover a finite refutation. Conversely, the cofinal and parity construction just checked supplies all new power-base and horizon witnesses,

and the retained reverse CRT theorem supplies all other witnesses. Thus this companion has the same existence interpretation, but by its own complete source route. It is not claimed to be a coordinate bijection with the shared-error source, and it does not impose $F = t!$. No least-possible evaluation point or global coefficient optimum is asserted by Lemma 3.3.

3.11 Literal selected equations and finite counts

The principal `signed/balanced238/equation.json` has 238 variables and 175 quadratic residuals. Its full expansion has 5236 terms and height 3968. The companion `signed/dimension237` has 237 variables, 174 rows, 18569 terms and height 7680. The factorial-free `signed/power239` has 239 variables, 175 rows, 5137 terms and height 3968. Each has exact degree four and every declared coordinate occurs. These are measured on complete coefficient arrays, not predicted from a count of proposed eliminations.

The balanced principal uses six unit projections after its arithmetic source: three exposed affine coordinates and the quadratic coordinates with historical labels `mul.281`, `add.292` and `mul.194`. The dimension companion also projects the affine coordinate `add.380`. The power companion applies the same six valid projections to its own distinct source. Every stage records its literal unit equation, coordinate order and inverse. Technical lifting sources `exposed241` and `power_exposed242` are retained for source binding and incidence construction; they are not additional headline improvements.

The main principal improves all four varying measures over the latest incoming Shared factorial error balanced equation $(241, 178, 5351, 15107)$ and its lower-height companion $(241, 178, 9544, 4512)$. Compared with that input's dimension point $(240, 177, 21482, 6144)$ it also improves every displayed measure. Against Round 19's $(244, 176, 5212, 3968)$ balanced point it trades 24 additional terms for six fewer variables and one fewer row. The power companion saves 99 terms relative to the new principal at the price of one more variable. No total ordering of these objectives is assumed.

4 Height one quartics with anchored equality

The source compiler acts on a finite quadratic system over unrestricted integers. Round 18 supplied certified normalization, row-sign balancing and opposite-occurrence fusion: equal monomials with opposite signs become a single incidence edge between their source rows. Round 19 additionally polarized square aliases and removed equality dummies. The present transformations absorb equality dummies into anchored products and contract selected internal flow edges. We first give the retained energy and witness argument, then prove both new steps. Every height claim concerns the entire collected polynomial, including all equality penalties.

4.1 Certified normalization and row orientations

Let the given source be $R_r(x) = 0$. We first use the preceding exact affine refactorization, when applicable: if the homogeneous quadratic part of a row is $\eta L(x)^2$, with $\eta = \pm 1$, introduce $z = L(x) + k$ and replace the row by $\eta z^2 + R_r(x) - \eta(L(x) + k)^2 = 0$. The new definition is monic and its value is uniquely determined. Shared affine definitions can be reused. Coefficient-by-coefficient substitution verifies every transformed row. The provided original source, transformed source and factor certificate fix this operation completely.

Normalize integer coefficients by monic definitions of integer constants $t_c = c$ and scale variables $s_{c,i} = cX_i$. A constant is defined from already defined smaller constants as $t_c = t_a + t_b$ when $a + b = c$, using $t_1 = 1$. If no such pair is available, introduce the smaller binary intermediate and use $t_c = t_{\lfloor c/2 \rfloor} + t_{\lfloor c/2 \rfloor} + (c \bmod 2)$. Each added definition is triangular and monic. The delivered DAG prefers two already-needed constants; in particular it does not introduce a binary intermediate merely because a binary-only recipe would do so.

For each absolute quadratic coefficient $c > 1$, form a finite graph on the source coordinates, with an edge $\{i, j\}$ for every term cX_iX_j. A loop at i requires selecting i. Choose any vertex cover V_c of this graph, and introduce $s_{c,i} = t_cX_i$ only for $i \in V_c$. A term cX_iX_j is then represented by $s_{c,i}X_j$ if i was selected, or by $X_is_{c,j}$ otherwise. The cover condition proves that one option always exists. The retained deterministic greedy cover starts with the loop vertices and repeatedly selects a highest-degree remaining vertex; optimality is not assumed. Constants and linear terms use t_c and t_cX_i, respectively. Unit coefficients need no constructor.

Splitting repeated summands into separate occurrences yields

$$S_r(y) = \sum_j \epsilon_{rj} m_{rj}(y), \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

Every normalized coordinate has a unique integral value over an original tuple. The checker verifies all triangular pivots and each exact source-row pullback, rather than trusting the constant-chain or cover-selection recipe. Thus normalization preserves both witness maps. The coefficient-factor choice, just like the later graph, is fixed by the certificate and is not a parameter of the polynomial.

Choose a fixed sign $\sigma_r \in \{-1, 1\}$ for each normalized row, and replace S_r by $T_r = \sigma_r S_r$. This preserves both its zero set and uniqueness of all constructors. The choice is part of the finite certificate, not a parameter of the final polynomial. The supplied heuristic flips a row when doing so reduces

$$\sum_m \left| \sum_{r,j:m_{rj}=m} \sigma_r \epsilon_{rj} \right|.$$

This balances opposite occurrences of the same logical monomial. Correctness does not require optimality of this heuristic.

4.2 Direct edges for cancelling occurrences

Create disjoint vertex blocks B_r for the rows and a hub block B_0. Inside each block choose a tree, with one independent unrestricted integer flow variable on each tree edge. A normalized occurrence is assigned to a vertex in its own row block. Partition the occurrences into sets of size one or two. A pair is allowed only when its occurrences are in distinct rows, have the same logical monomial, and have opposite signs in T_r.

A singleton of sign ϵ is represented by an edge from its row vertex to a hub vertex carrying ϵM. A pair $+m$ in row r and $-m$ in row s is represented by one edge from its row-r vertex to its row-s vertex, carrying M. Here M is a degree-two homogeneous monomial in aliases of the logical factors of m: a linear term is multiplied by a unit alias and a constant uses two unit aliases. Each factor alias has one declared logical value. The two occurrences of a pair use *the same* edge and the same factor aliases.

The graph is required to be simple. The constructor rejects a pairing if its proposed unordered vertex pair was already used; remaining hub edges are placed without parallel edges. The certificate lists every bound source occurrence, and the checker verifies that each appears exactly once. More precisely, the edge boundary at the row-block level equals its list of signed source occurrences. Consequently, after substituting the intended logical values,

$$\sum_{v \in B_r} b_v = T_r(y), \qquad \sum_{v \in B_0} b_v = -\sum_r T_r(y). \tag{20}$$

where b_v is the divergence from prescribed edges alone. A fused edge contributes $+M$ and $-M$ to precisely its two bound rows; hence (20) is unchanged by fusion. This is the inherited occurrence-boundary identity.

4.3 The coefficient-height-one energy

Let d_v denote the outgoing-minus-incoming sum of all prescribed and free edges, and put

$$E = \frac{1}{2}\sum_v d_v^2.$$

Each edge has two endpoints, so its diagonal energy coefficient is one. Two distinct edges of a simple graph meet at most once, so their formal cross coefficient is 0 or ± 1. Give each prescribed product an initial private factor alias, and share its other logical factor as in the previous incidence compiler. The initial fully expanded energy has height one. Subsequently identify only aliases of the same logical value, and accept an identification only when full exact coefficient collection still has coefficients in $\{-1, 0, 1\}$. Such substitution preserves the sum-of-squares identity even when signed collisions cancel.

For every remaining non-unit logical class with at least three aliases use

$$C(a_1, \dots, a_k) = \sum_i a_i^2 - \sum_i a_i a_{i+1}, \quad a_{k+1} = a_1.$$

Since $2C = \sum_i (a_i - a_{i+1})^2$, it is nonnegative and vanishes exactly when all aliases agree. A singleton needs no penalty. A class with two aliases is either completed to a three-cycle by one dummy, or uses

$$Q(a, b) = a^4 + b^4 - a^3 b - ab^3 = (a - b)^2(a^2 + ab + b^2).$$

This is nonnegative and vanishes exactly at $a = b$; also $2Q = (a^2 - b^2)^2 + (a^2 - ab)^2 + (ab - b^2)^2$. The latter option is used only when neither alias occurs as a prescribed square, which ensures its four monomials cannot collide with the incidence energy. For all unit aliases, adding dummies if necessary so that $\ell \geq 2$, put

$$U = 1 + \sum_{i=1}^{\ell} u_i^2 - u_1 - u_\ell - \sum_{i=1}^{\ell-1} u_i u_{i+1}.$$

Twice U is $(u_1 - 1)^2 + \sum_{i<\ell}(u_i - u_{i+1})^2 + (u_\ell - 1)^2$. Thus the final literal polynomial is

$$\boxed{P = E + \sum C + \sum Q + U.} \tag{21}$$

It is nonnegative over the reals. All monomials have total degree at most four. The full coefficient files are independently checked to have exact degree four, every nonzero coefficient equal to $+1$ or -1, constant coefficient one, and every stated variable present. The retained square-alias transformation below preserves this energy identity and revises a selected equality penalty with its exact zero set intact.

4.4 Forward map, inverse map, and uniqueness

Given a source tuple, evaluate the monic affine, constant and scale constructors, assign each ordinary alias its logical value and every unit alias the value one. Assign equality dummies the value of their class. All equality and unit penalties then vanish.

Root each internal tree at its first listed vertex. For a tree edge with child v, let $\tau_v = \pm 1$ be its incidence sign at the child, and set

$$w_v = -\tau_v \sum_{t \text{ in the subtree of } v} b_t.$$

This is an explicit integral polynomial in the source coordinates. Eliminating leaves proves that all nonroot divergences vanish. Formula (20) gives root divergence T_r in each row block and $-\sum_r T_r$ in the hub block. Thus the canonical lift satisfies, universally,

$$P(L(x)) = \frac{1}{2}\left(\sum_r R_r(x)^2 + \left(\sum_r \delta_r R_r(x)\right)^2\right). \tag{22}$$

where δ_r is the certified sign of the normalized row descending from the original residual R_r. Constructor rows pull back to zero. In particular every source zero has an integer lift. The affine-definition rows also vanish identically under their canonical substitution, so (22) composes unchanged through affine refactorization. One must not replace (22) by a bare sum of the original squares.

Conversely, if $P(z) = 0$, nonnegativity in (21) forces every equality penalty and U to vanish. Hence all ordinary aliases in a class agree, and all unit aliases equal one. Also $E = 0$ forces every vertex divergence to vanish. Summing them over a row block cancels its internal tree flows; by the certified occurrence-boundary identity the result is precisely $T_r(y) = 0$. Since $\sigma_r = \pm 1$, all normalized rows $S_r(y)$ vanish. Triangular monic constructors then force the constant, scale and affine coordinates to their intended values. Reading the original logical coordinates yields an original source zero.

The same argument proves uniqueness: constructors, aliases, units and dummies have unique values over the original tuple, and leaf elimination uniquely determines every internal tree flow. Thus the displayed forward map and coordinate projection are inverse bijections between integer zero sets. No sign restriction on a final variable, division by a variable, or unproved boundedness assertion is used.

4.5 Square polarization removes equality dummies

Suppose a logical value currently has exactly two factor aliases a, b, and its equality penalty uses a fresh dummy d:

$$C(a, b, d) = a^2 + b^2 + d^2 - ab - bd - da.$$

The reason for retaining this dummy can be a prescribed square a^2 inside the incidence graph: directly substituting the two-alias penalty $Q(a, b)$ may collide with its degree-four diagonal contribution. The retained Round 19 operation first changes selected prescribed squares into products of same-logical aliases, $a^2 \mapsto ab$ (or $b^2 \mapsto ab$). It then replaces $C(a, b, d)$ by $Q(a, b)$ and deletes the dummy coordinate d.

Theorem 4.1. *Make any finite selection of such same-logical prescribed-square changes, and replace each affected two-alias equality cycle by Q. If the full collected final polynomial has coefficient height one, it is a height-one quartic with an integer zero set bijective to the original arithmetic source. Each deleted equality dummy reduces its dimension by one.*

Proof. Changing a prescribed edge's label from a^2 to ab leaves its two incidence endpoints and orientation unchanged. The graph energy therefore remains exactly $\frac{1}{2}\sum_v d_v^2$, with its divergences evaluated using the modified labels. It is still nonnegative, and all labels still have degree at most two. Also

$$Q(a, b) = (a - b)^2(a^2 + ab + b^2) \geq 0,$$

and $Q(a, b) = 0$ if and only if $a = b$: the second factor equals $(a + b/2)^2 + 3b^2/4$ and vanishes only at $a = b = 0$.

At any zero of the final energy, every nonnegative equality term is zero, so same-logical aliases agree and units equal one. Consequently every polarized label ab has exactly the value previously specified by a^2. The occurrence-boundary identity is unchanged after logical substitution. Vanishing vertex divergences therefore give all normalized source rows, and monic reconstruction gives an actual arithmetic source zero.

Conversely, for an original source zero assign both a and b their common logical value. All polarized edge labels then have their old values, all new Q terms vanish, and the same unique subtree flow assignment makes every vertex divergence zero. Deleted dummies have unique old value $d = a = b$, so extension and deletion are inverse on the zero sets. This also proves that the source-polynomial lift identity (22) survives universally, not only at source zeros.

The degree is at most four by construction. Exact degree four and coefficient height one are finite properties of the entire collected polynomial and are explicitly checked. They do not follow

merely from the simplicity of the incidence graph after monomial substitution. The operation is accepted only after this full collection check, including all equality and unit terms. Thus possible collisions do not invalidate the claim. □

The theorem makes no assertion that an arbitrary polarization is safe for height one, nor that a chosen finite sequence is optimal. It separates the unconditional witness proof from the coefficient test selecting a particular literal endpoint.

4.6 Absorbing equality dummies into anchored products

A different new operation removes an equality dummy without changing any prescribed square label. Suppose two aliases a, b of one non-unit logical value are tied by a fresh dummy d through

$$C(a, b, d) = a^2 + b^2 + d^2 - ab - bd - da.$$

Assume d occurs nowhere else. Let u be an existing unit alias already anchored to one by U. Substitute $d = au$ and delete the coordinate d. The new equality penalty is

$$\widetilde{C}(a, b, u) = a^2 + b^2 + a^2u^2 - ab - abu - a^2u. \tag{23}$$

It has degree four and is the literal polynomial $C(a, b, au)$.

Theorem 4.2. *Under the stated incidence restrictions, replacing $C(a, b, d)$ by (23) preserves the integer zero set by a polynomial bijection after deleting one coordinate. It also preserves the universal source lift identity (22). If the entire resulting collected polynomial has height one, exact degree four and all declared coordinates present, it is a certified height-one quartic with one fewer independent integer variable.*

Proof. For arbitrary real a, b, u,

$$2\widetilde{C} = (a - b)^2 + (b - au)^2 + (au - a)^2 \geq 0.$$

At a zero of the complete energy, its unchanged unit penalty forces $u = 1$. The new equality penalty then vanishes exactly when $a = b$. Thus the old dummy has the unique value $d = a = b$, while every other summand is unchanged. Extending with $d = a$ maps a new zero to an old zero. Conversely, at an old zero $d = a = b$ and $u = 1$, so deleting d gives a new zero. These maps are inverse polynomial maps on the zero sets.

For a canonical source lift, all unit aliases equal one and all ordinary aliases equal their logical values. Consequently $au = a = d$ identically on that lift, including away from source zeros. No prescribed incidence label or divergence has changed, so the universal lift identity is retained.

Polynomial substitution preserves nonnegativity but not automatically coefficient height. The term a^2u^2 may collide with graph-energy terms, and several absorbed dummies may share a unit alias. The acceptance test therefore collects the whole energy after every proposed substitution, including all other penalties. It explicitly checks the finite claims about coefficients, degree and the live coordinate set. No extra unit coordinate is introduced: u was already present and anchored. □

Several such absorptions can be composed, using the same or different existing unit aliases, provided the full polynomial test accepts the frozen final choices. Unlike Round 19 polarization, this operation leaves the prescribed square monomials in the incidence graph untouched. Unlike a formal substitution $d = a$ alone, it does not insert a coefficient -2 into the equality row before the complete energy is considered. Its quartic product with an already anchored unit provides additional distinct monomials that can retain height one.

4.7 Contracting an internal flow edge

The new operation changes the divergence energy rather than only the alias penalties. It removes a flow coordinate on an internal tree edge and identifies its endpoints. It is therefore not an alias identification or an instance of square polarization. Its soundness is unconditional; coefficient height one is a separate exact acceptance test.

Theorem 4.3. *Let an incidence presentation have the energy form* (21)*, with every internal edge belonging to a tree inside one row block or the hub block. Choose an internal edge from u to v, with its own flow coordinate w. Delete that edge, identify its vertices, and replace their two divergences by their sum. Retain all prescribed occurrence labels and equality and unit penalties. Then the old and new integer zero sets are in polynomial bijection, preserving the original arithmetic coordinates. The new presentation has one fewer flow coordinate and degree at most four. If full collection of its polynomial has height one, exact degree four, and every listed variable occurs, it is a certified height-one quartic on that smaller coordinate set.*

Proof. Write the old two divergences as

$$d_u = a_u + \epsilon w, \qquad d_v = a_v - \epsilon w, \qquad \epsilon \in \{-1, 1\},$$

where a_u, a_v are integer polynomials independent of w. No other divergence or penalty depends on w, because it is the independent flow coordinate of this one edge. The merged divergence is $d_{uv} = a_u + a_v$. Thus the new polynomial is exactly

$$P' = \frac{1}{2}\left((a_u + a_v)^2 + \sum_{t \neq u,v} d_t^2\right) + \sum C + \sum Q + U.$$

This expression proves nonnegativity and the degree bound without any claim that the old and new polynomials themselves agree.

At an old zero, nonnegativity forces $d_u = d_v = 0$ and every other summand to vanish. Deleting w therefore gives $a_u + a_v = 0$ and a new zero. Conversely, at a new zero all its displayed summands vanish. Set

$$w = \epsilon a_v.$$

Then $d_v = 0$, while $d_u = a_u + a_v = 0$; all other old summands are unchanged. Indeed the stronger identity $P|_{w=\epsilon a_v} = P'$ holds for every remaining tuple: the two old divergences become $a_u + a_v$ and zero. This is an integral polynomial restoration, and it is unique on zeros because ϵ is a unit. Restoration and deletion are mutually inverse on the zero sets. All equality and unit arguments, and hence all recovered logical coordinates, are identical on the two sides.

The edge's endpoints belong to one block. Summing the contracted vertex divergences in that block therefore gives the same boundary sum as before contraction. In particular (20) continues to recover exactly the same normalized source row; no source occurrence is deleted. Contracting an edge of a tree leaves a tree. The remaining flows are still uniquely recovered by leaf elimination, or equivalently by composing the one-edge restoration above.

Finally, degrees and coefficient heights concern the polynomial after all monomial collisions, including equality and unit terms. Contracting vertices can create parallel prescribed incidences, so the initial simple graph argument alone is no longer a sufficient height certificate. The selection test reconstructs and collects the entire new energy and checks its actual integer coefficients. This establishes the stated height, exact degree and coordinate census only for the accepted finite instance. □

Corollary 4.4. *Any finite certified sequence of accepted internal-edge contractions preserves the calibrated integer zero set. Its inverse is obtained by restoring the deleted flows in reverse order. If all k deleted flows were distinct live coordinates and no other coordinate is removed, its independent-variable count decreases by exactly k.*

Proof. Compose the polynomial deletion and restoration maps of Theorem 4.3. Every step preserves the block boundary identity and leaves internal blocks as trees. The final independently collected array fixes the actual coordinate count and all measured polynomial invariants. □

This argument does not predict which edges pass the height test or minimize the support. A contraction can save a variable while creating more monomials; it can also fail the height test entirely. Adaptive selection is a construction method, not a proof of optimality. The released certificate freezes each accepted edge, so reproduction requires no heuristic search.

4.8 Selected literal outputs and matched sources

Endpoint	Variables	Terms	Original arithmetic source
`minimum503`	503	24193	New exposed 241
`compact505`	505	20491	New power exposed 242
`compact512`	512	18255	New power exposed 242
`balanced557`	557	11999	Retained grouped 263
`sparse617`	617	8318	Retained sparse 285
`sparser713`	713	7176	Retained sparse 285

All six have exact degree four, coefficient height one, constant coefficient one, no equality dummy coordinates, and every declared unrestricted coordinate present. Each supplies its entire coefficient array, literal polynomial text, graph certificate and source binding. The 503-variable point is a dimension tradeoff: the 505-variable point has much smaller support. The latter improves both variables and support over Round 19's 528-variable point. The 512-variable choice similarly dominates the old 534-variable point; the remaining three dominate their respective old 580-, 620- and 717-variable counterparts.

The first source has 241 variables and 178 rows before a fixed 26-coordinate affine refactorization; it then has 267 variables and 204 rows, and normalization gives 339 variables and 276 rows. Its final graph uses 16 initial hub vertices and 21 remaining flows. The next two use the separate power-base source with 242 variables and 178 rows, then 269 variables and 205 rows after 27 affine definitions; normalization gives 343 variables and 279 rows. Their initial hub counts are 20 and 24, and their final flow counts are 22 and 26. None is identified with the literal 238-variable balanced equation merely because it descends from the same shared-error arithmetic.

To isolate the structural changes from new arithmetic sources, five matched replays authenticate the exact old graph and polynomial hashes before applying the new operations:

Matched Round 19 graph	Before variables/terms	After variables/terms
Minimum selected	528/20816	508/20808
Compact	534/18543	518/18543
Balanced	580/12001	557/11999
Sparse	620/8318	617/8318
Sparser	717/7176	713/7176

The last four use anchored equality absorption alone. In the first, absorption alone gives 511 variables and 20816 terms; internal flow contraction and renewed same-logical alias contraction give 508 and 20808. Thus each matched comparison improves dimension without increasing support. The 508- and 518-variable controls are reproduced evidence, not additional released endpoints. The final selected source and graph choices are frozen in the catalog; their quality does not depend on claiming an exhaustive search or an optimality theorem.

5 Polynomial centers and eleven integer equations

5.1 Complete polynomial definitions

The independent coordinates are exactly

$$(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}.$$

The six fixed ports are the authentic source34 numerals specified in Section 2; none is another variable. Define the following ordinary polynomial abbreviations:

$$\begin{aligned} J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4, \\ X &= qW, & Y &= v, & E &= XY, \\ a &= E + Y, & A &= a + 2, & H &= 4a + 3, \\ \Delta &= a^2 + H = A^2 - 1, \end{aligned}$$

$$\begin{aligned} F &= (K_0 + W)C - z(q - 1), \\ R &= [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J, \\ k &= R + 1 + hE, \qquad \eta = c - Yk, \\ D_{\mathrm{err}} &= (\mathrm{cone_A}F - q)^2 + (\mathrm{cone_B}C - q)^2, \\ B_0 &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\ \mathcal{B} &= Y^2 B_0 - k^2 q^8, \qquad D = \mathcal{B} - 4, \\ \ell &= XY^2, \qquad D_t = 1 + \ell(\ell + 1)k^2, \\ D_g &= 1 + DD_t s^2, \qquad P = D_t D_g. \end{aligned}$$

The cofactor, auxiliary norm and carrier registers are

$$\begin{aligned} D_m &= 1 + \Delta c^2, & B &= rAD_m, & U &= rc^2 D_m, \\ I_2 &= c^4 B = Ac^2 U, & D_f &= 1 + \Delta I_2, & Q &= \Delta^2 I_2, \\ \delta &= Q - 1, & V &= D_f(c - R + 4AUT) - c, & C_y &= QV^2 - 1, \\ K &= X(3c - 2X) + 2, & M &= 2q^3, & D_c &= HM, \\ S &= srHM^5, \end{aligned}$$

$$z_c = D_c T + MK + H(v - q^3), \qquad A_c = (z_c^2 + 3D_c^2)(z_c^3 - 19D_c^3).$$

The following inherited Round 18 centers fix the comparison boundary:

$$\begin{aligned} \Psi_{316} &= [(8\Delta + 4)U - rc^4]V, & L_{316} &= 8, \\ \Psi_{308} &= \Psi_{316} + 2U^3 T(c^2 - 2), & L_{308} &= 8, \\ \Psi_{304} &= \Psi_{316} + U^2 T[2U(c^2 - 2) + 16A^3], & L_{304} &= 8, \\ \Psi_{300} &= [(32\Delta + 16)U - 4rc^4]V \\ &\quad + U^2 T[8U(c^2 - 2) + 64A^3 + 3rc^6], & L_{300} &= 32. \end{aligned}$$

Every center is an integer polynomial independent of n. For any selected even weight L and integer center Ψ, put

$$\begin{aligned} Z &= S(n + \Psi) - A_c, \\ p_a &= \delta Z^2, \qquad p_b = \delta S^2 P, \qquad p_c = L^2 S^2 B D_f C_y, \\ \mathcal{N}_{L,\Psi} &= (p_a + p_b - p_c)^2 - 4p_a p_b. \end{aligned} \tag{24}$$

The equation is $\mathcal{N}_{L,\Psi} = 0$ over $\mathbb{Z}^{11}$. The displayed centers are retained comparison controls defining the old frame. The new shorter centers and the separately rebased degree-298 frame are specified below. No division, radical or variable exponent is an operation of this polynomial. The literal graph shares the identities

$$P = D_t + (\mathcal{B} - 4)(D_t s)^2,$$

$$\mathcal{B} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^3)^2], \qquad A^2 = \Delta + 1.$$

Sharing these registers changes graph cost, not the polynomial.

5.2 An even weight and integral center lemma for the old frame

Theorem 5.1. *For every fixed nonzero even L and every integer polynomial Ψ in the ten coordinates other than n, the projection of the integer zero set of (24) onto those ten coordinates equals that of the retained Round 17 free-scale equation. The latter uses $L = 2$ and $\Psi = (2\Delta + 1)UV$.*

Proof. We first record the unconditional arithmetic facts from Sections 2–4 of the retained `LINEAR_COFACTOR_ROUND17_PROOF.md`. They do not assume a zero of the present norm. The authentic affine restrictions give

$$\begin{aligned} q &\equiv 0 \pmod 2, & q &\equiv 2 \pmod 3, & |q| &\geq 16, \\ X &\equiv 0 \pmod 8, & X &\equiv 2 \pmod 3, & |X| &\geq 4|q|. \end{aligned}$$

Thus $A \neq 0$, $\Delta = A^2 - 1 > 1$, $\Delta \equiv 0 \pmod 3$, H is odd and nonzero, and $D_m, D_t \geq 1$. The exact modulo-sixteen budget identity gives $D \equiv 8$ or $12 \pmod{16}$, so D is nonzero and nonsquare. Also $P \equiv 1 \pmod 4$ and $P \neq 0$. Both D_f and δ are nonzero, Q is even, and δ is odd.

The rational polynomial $\pi(t) = (t^2 + 3)(t^3 - 19)$ has no rational root. Since $D_c \neq 0$,

$$A_c = D_c^5 \pi(z_c/D_c) \neq 0.$$

If $S = 0$, the output norm is $(\delta A_c^2)^2 \neq 0$. Every new zero therefore has $S, s, r, B \neq 0$.

Put $D_a = \delta C_y$ and $C_* = BD_f D_a$. One has $D_a > 0$: if $c = 0$, then $Q = 0$ and $D_a = 1$; if $c \neq 0$ and $B < 0$, then $\delta, C_y < 0$; if $c \neq 0$ and $B > 0$, then $D_f > |c| + 1$, so $V = D_f b - c$ is nonzero for every integer b, and $\delta, C_y > 0$. Moreover $BD_f > 0$ whenever $c \neq 0$. Hence $C_* \neq 0$, and its only possible negative case is $c = 0, B < 0$.

The four-conjugate norm identity for (24), after normalizing by nonzero S and multiplying by δ, gives a rational signed sum

$$\pm L\sqrt{C_*} \ \pm \delta\sqrt{P} \in \mathbb{Q}. \tag{25}$$

If exactly one radicand is negative, its nonzero imaginary contribution cannot cancel. If both are negative, then $c = 0$, $\delta = -1$ and $C_* = B$. Cancellation would force $L^2 B = P$, impossible modulo four because L is even and $P \equiv 1 \pmod 4$. Therefore $C_*, P > 0$.

Write these positive integers as squares times positive squarefree integers. Different nontrivial squarefree radicals are independent over $\mathbb{Q}$. The only remaining possible irrational cancellation would have the same nontrivial squarefree part in both terms of (25). Its first coefficient would be even, whereas its second coefficient would be odd: L is even and δ, P are odd. Cancellation is impossible. Thus C_* and P are positive integer squares.

The cofactor coprimalities are unchanged:

$$\gcd(B, D_f) = \gcd(B, D_a) = \gcd(D_f, D_a) = 1.$$

Indeed $D_f \equiv 1 \pmod B$ and $Q \equiv 0 \pmod B$, giving $D_a \equiv 1 \pmod B$. Modulo D_f, one has

$$Q \equiv -\Delta, \quad \delta \equiv -A^2, \quad V \equiv -c, \quad C_y \equiv -D_m, \quad D_a \equiv A^2 D_m.$$

Since $B = rAD_m$, while $D_f \equiv 1$ modulo both A and D_m, these imply the last coprimality. Therefore $|B|, |D_f|, D_a$ are squares. As $D_a > 0$, B, D_f have the same sign. But $D_f \equiv 1 \pmod 3$ excludes a negative square, so there are positive integers u, f, e with

$$B = u^2, \qquad D_f = f^2, \qquad D_a = e^2. \tag{26}$$

Similarly $\gcd(D_t, D_g) = 1$ and $D_t \geq 1$ give $D_t = \tau^2$, $D_g = g^2$ with $\tau, g > 0$. Since $s \neq 0$, the guard forces $\mathcal{B} \geq 4$, strengthened by its modulo-sixteen class to $\mathcal{B} \geq 12$. It follows that $k, Y \neq 0$, $B_0 > 0$, and

$$Y^2 B_0 > k^2 q^8, \qquad B_0 \leq k^2 q^2, \qquad |Y| > |q|^3.$$

The centered strip from $B_0 > 0$ then excludes $c = 0$.

Define proof-only integers $I = c^2 u$ and $T_{\text{old}} = uT$. By (26),

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad V = f^2(c - R + 4IT_{\text{old}}) - c.$$

The rational number $y = e/\delta$ satisfies

$$QV^2 - (Q-1)y^2 = 1. \tag{27}$$

It is integral by adjacent-coefficient descent. If b is its reduced denominator, then $b^2 \mid Q - 1$. After choosing signs, for $|V| > 1$ the unimodular transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves (27) and the reduced denominator, with $0 < |V'| < |V|$. The first coordinate is integral because $(Q-1)y$ is integral; denominator preservation also follows from $\gcd(2Q-1, b) = 1$. For $V > 1$ and $Q > 1$, the norm gives

$$V < y < \frac{(2Q-1)V}{2(Q-1)}.$$

The squared upper bound minus y^2, multiplied by $4(Q-1)^2$, is $V^2 + 4(Q-1) > 0$. Consequently $0 < V' < V$. Descent reaches $|V| = 1$, where $|y| = 1$, forcing $b = 1$. The formal $c = 0$ case would directly give $y = -1$, although it is already excluded here.

The two normalized roots of the original four-conjugate equation are now integers $Lufy$ and τg. Hence Z/S is integral. Since Ψ is integral, $S \mid A_c$. No size or sign condition on Ψ was used. Keep the ten outer coordinates of the new zero and set

$$n_{\text{old}} = A_c/S + 2ufy + \tau g - (2\Delta + 1)UV.$$

This is an integer and makes one old conjugate factor vanish, producing an actual Round 17 zero. Conversely an old zero supplies these same integral roots and divisibility by the retained theorem. Keeping its ten outer coordinates, set

$$n_{\text{new}} = A_c/S + Lufy + \tau g - \Psi.$$

One new factor vanishes. These constructions prove equality of projected zero sets. They need not be mutually inverse on n, since the norm can have several sign branches. For a fixed L, changing only the center is separately a literal bijective integral shear of n. □

5.3 The complete inherited calibration chain

Theorem 5.1 reduces both directions to an actual old zero, not merely a list of necessary Pell properties. Thus its inherited scale, sign, rank and syntactic interfaces remain exactly applicable. For clarity, the retained source proof proceeds as follows.

Carrier divisibility first gives $M \mid H(v-q^3)$, even Y, and

$$H \mid q^{15}K_q^5, \qquad K_q = X(X+2ac)-1.$$

The strong and auxiliary pairs, together with $c^2 \mid I$, recover the main Pell root by absolute folding. With $D_t = \tau^2$, the centered strip, $|Y| > |q|^3$ and the authentic packing inequalities

$$|q|(|q|^2-1) < R < |q|^4 - 2,$$

the signed rank theorem yields

$$c = \operatorname{sgn}(c)\psi_A(R), \qquad R = 2t+1, \qquad |H| > |X|(|X|-3)^t.$$

The powered projection at the actual large base forces $\operatorname{sgn}(c)X = 2^R$. Reduction modulo three fixes $c > 0$ and $X = 2^R$; the old strip fixes $k, Y > 0$. Exact $H \mid K$ restores $Y = (2v_{\mathrm{old}}+1)q^3$. The uniform half-binomial theorem then restores the positive native interval, and fresh odd strong witnesses produce an actual signed Q41 source zero. Only that restored zero is passed to the retained source34/native/compiler/finite-refutation calibration.

Conversely, canonical source witnesses give $\mathcal{B} > 64$. After choosing the guard pair s, g, the old proof chooses one strong pair I, f with $4sAdc^2 \mid I$, where $D_m = d^2$, and sets

$$u = I/c^2, \qquad t_{\mathrm{strong}} = I/(Adc^2), \qquad r = At_{\mathrm{strong}}^2.$$

That pair is kept fixed. The supported-modulus theorem interpolates T_{old} modulo usr using the same pair, while an intersective root of π modulo sr makes A_c/S integral. There is no reselection of the strong pair after a residue request. For changes confined to this old frame, the center modification replaces only its final integer n by n_{new} above and introduces no new interpolation or choice-order assumption. Full inherited proofs and their source identities are retained separately; they are not promoted here to a newly checked formal-kernel endpoint.

5.4 The incoming rebased frame and two shorter centers

The latest Nearest factorial input already changes the first-index congruence and the magnitude budget together, obtaining degree 298 at 150 gates. Its rebased frame replaces the two old definitions by

$$k = R + 1 + h\ell, \qquad \mathcal{B}_6 = Y^2B_0 - k^2q^6, \qquad \ell = XY^2, \tag{28}$$

and uses $D = \mathcal{B}_6 - 4$. Every other register in the preceding complete definitions is retained. The older frame instead keeps $k = R + 1 + hE$ and $\mathcal{B}_8 = Y^2B_0 - k^2q^8$. These are different polynomial families. A degree-298 endpoint in the rebased family does not contradict a degree-300 bound in the older one.

Set $\gamma = 8\Delta + 4$ and $\lambda = 4(\gamma U - rc^4)$. The new centers are

$$\Psi'_{298} = \lambda V + U^2T\left\{rc^4[(\gamma-1)(c^2-2)-2] + 64A\Delta\right\}, \tag{29}$$

$$\Psi'_{300} = \lambda V + U^2T\left\{rc^4[(\gamma-1)(c^2-2)] + 64A\Delta\right\}. \tag{30}$$

Use Ψ'_{298} with the rebased front and Ψ'_{300} with the old front, in both cases with $L = 32$ in (24). The ten outer coordinates and the eleventh coordinate n remain unrestricted integers. The six authentic numerical ports are unchanged.

Theorem 5.2. *The two displayed equations have polynomially bijective integer zero sets with their respective calibrated parents. They have exact degree and live binary-operation counts* 298/147 *and* 300/145*, respectively.*

Proof of the zero set assertion. The incoming rebased center is

$$\Psi_{298}^{\mathrm{p}} = [(32\Delta + 16)U - 4rc^4]V + U^2T[8U(c^2 - 2) + 64A^3 + rc^4(3c^2 - 16)].$$

The old degree-300 center replaces its final term $rc^4(3c^2 - 16)$ by $3rc^6$. The exact identities $U = rc^2 + \Delta rc^4$ and $A^3 = A\Delta + A$ give

$$\Psi'_{298} - \Psi_{298}^{\mathrm{p}} = U^2T(16rc^2 - 64A) =: \epsilon_{298},$$
$$\Psi'_{300} - \Psi_{300}^{\mathrm{p}} = U^2T(16rc^2 - 64A - 14rc^4) =: \epsilon_{300}.$$

Each ϵ is an integer polynomial independent of n. Keeping all outer coordinates and setting $n_{\mathrm{p}} = n + \epsilon$ makes the parent's Z exactly equal to the new Z, hence makes the entire parent and new polynomials equal under substitution. The inverse is $n = n_{\mathrm{p}} - \epsilon$. This is a bijection on the full integer zero sets, not a rational map or a test at sample points. It proves the new calibration once each respective parent is calibrated. The operation and exact degree assertions are proved below. □

5.5 The rebased parent keeps its guard and its late scale proof

The changed frame in (28) must not be confused with an earlier rejected guard deletion. The complete incoming ordinary proof is retained with its dependencies. Its critical witness argument is stated here to fix the logical handoff precisely.

The unchanged even-weight norm, coprime-product and adjacent-coefficient descent recover $B = u^2$, $D_f = f^2$, $D_t = \tau^2$ and an integral auxiliary y, with $u, f, \tau > 0$, together with $\mathcal{B}_6 \geq 12$, $S \neq 0$ and $S \mid A_c$. Before extraction one still has $\Delta > 1$, odd nonzero H, $D \equiv 8$ or $12 \pmod{16}$, $P \equiv 1 \pmod 4$, and odd nonzero δ. The possible cancellation of two negative radical parts would require $1024B = P$, excluded modulo four. The exact coprimalities

$$\gcd(B, D_f) = \gcd(B, \delta C_y) = \gcd(D_f, \delta C_y) = 1, \qquad D_f \equiv 1 \pmod 3$$

therefore supply positive individual squares. With the proof-only values $I = c^2u$ and $T_{\mathrm{old}} = uT$, the recovered equations are

$$f^2 - \Delta I^2 = 1, \qquad V = f^2(c - R + 4IT_{\mathrm{old}}) - c,$$
$$(\Delta I)^2V^2 - [(\Delta I)^2 - 1]y^2 = 1.$$

In particular $c^2 \mid I$. The denominator descent uses this integral quantity, not an assumed integral value of a rational square root.

The guard $\mathcal{B}_6 > 0$ gives $kY \neq 0$, $B_0 > 0$ and $|Y| > |q|^2$. Its centered disk supplies the common nonzero signs of $q, F, C, q - F - C$, the strip $|c/k - Y| < 1/2$, and the authentic packing bound

$$|q|(|q|^2 - 1) < R < |q|^4 - 2.$$

Since $|X| \geq 4|q|$ and $|q| \geq 16$,

$$|\ell| = |X|Y^2 > 4|q|^5 > 3R + 1.$$

The carrier valuation gives $M \mid H(Y - q^3)$, whose 2-primary part makes Y even, and $H \mid q^{15}K_q^5$ for $K_q = X(X + 2ac) - 1$. It does *not* yet give $q^3 \mid Y$.

The first Pell recurrence supplies $\psi_{2\ell+1}(m) \equiv m \pmod{2\ell}$. Together with $k = R + 1 + h\ell$ it gives the signed congruence $\varepsilon\, 2m \equiv R + 1 \pmod{|\ell|}$. Its positive branch forces $m \geq (R + 1)/2$; the negative branch forces $m > R$. The retained absolute root folding and nonodd strong-rank lemma, using $c^2 \mid I$ rather than an unjustified oddness assumption, and the two auxiliary residues give the main index $p = R$. The negative branch is impossible, and no wrap in the positive branch

gives $2m = R + 1$. The retained ratio argument needs $|Y| \geq 6$, which has already been proved; it does not need the as-yet unavailable cubic scale. Writing $R = 2t + 1$, it yields

$$|Y| > (|X| - 3)^t/2 - 1, \qquad |H| > |X|(|X| - 3)^t.$$

Put $x = |X|$ and $\Pi = (\operatorname{sgn}(c)X - 2^R)(2^R \operatorname{sgn}(c)X + 1)$. The inherited whole-product bound, applied with these proved hypotheses, gives

$$H \mid q^{15}\Pi^5, \qquad |q|^{15}|\Pi|^5 < 2^{20t-10}x^{25}, \qquad |H| > x^{t+1}/2^t.$$

The authentic bounds $x \geq 2^{64}$ and $t \geq 36$ make $43t - 1526 \geq 22$, so the divisible integer has magnitude below $|H|$. Thus $\Pi = 0$ and $\operatorname{sgn}(c)X = 2^R$. Reduction modulo three fixes $c > 0$ and $X = 2^R$, and the surviving sign branch and strip give $k, Y > 0$. Only at this point does $q \mid X$ show that $|q|$ is a power of two. Now odd H is coprime to $M = 2q^3$, and the scale is recovered late:

$$Y = (2v_{\text{scale}} + 1)q^3, \qquad v_{\text{scale}} \in \mathbb{Z}.$$

The centered half-binomial theorem then gives $0 < \eta/k < 1/4$.

A general rebased zero need not satisfy $\mathcal{B}_8 > 0$. Soundness therefore returns to the earlier B_0-only source, not to a stronger free-scale guard. In that native handoff set $h_{\text{old}} = Yh$. Then $R + 1 + h_{\text{old}}XY = R + 1 + hXY^2 = k$ exactly. The retained construction regenerates the required fresh odd strong and auxiliary witnesses, restores its native interval and square coordinate, and passes an actual source zero to its compiler/refutation theorem. It does not assert that the current multiplier was already odd.

Conversely the canonical source has $k = 2\psi_{2\ell+1}((R + 1)/2)$, whose recurrence implies $4\ell \mid k - R - 1$, so the new h is integral. Its $\mathcal{B}_8 > 64$ implies $\mathcal{B}_6 > \mathcal{B}_8 > 64$. Choose the Pell guard pair, then one strong pair satisfying $4sAdc^2 \mid I$, then u, r, then an intersective-quintic residue. Interpolate T_{old} modulo usr with that same fixed pair; it is not reselected after imposing a residue. The retained divisibility proofs make

$$n = A_c/S + 32ufy + \tau g - \Psi$$

an integer and make one norm factor vanish. This completes the identified parent's witness construction. The exact center shear adds no further number-theoretic assumption or witness-order choice.

5.6 Exact degrees at the authentic coefficients

For an independently checkable cancellation calculation, work first in the free ring on (A, c, r, T, R) with weights $(3, 1, 1, 1, 5)$, set $\Delta = A^2 - 1$ and keep the literal U, B, D_f, Q, V. Define

$$D_\Psi = \Psi^2 - 1024BD_fV^2, \qquad E_\Psi = (Q - 1)D_\Psi - 1024BD_f(V^2 - 1).$$

Complete exact collection gives the following full top forms:

$$\begin{aligned}
[D_{\Psi'_{298}}]_{80} &= 512A^{15}c^{24}r^5TR - 8192A^{17}c^{22}r^5T^2,\\
[E_{\Psi'_{298}}]_{108} &= 512A^{22}c^{30}r^6TR - 24576A^{24}c^{28}r^6T^2,\\
[D_{\Psi'_{300}}]_{82} &= 512A^{16}c^{26}r^6T^2,\\
[E_{\Psi'_{300}}]_{110} &= 512A^{23}c^{32}r^7T^2.
\end{aligned}$$

All higher-weight coefficients vanish. The supplied independent checker reconstructs every coefficient, not only these displayed leading terms: the two remainder pairs have 449/881 and 455/888 terms, respectively.

Put $W_0 = Sn - A_c$. The exact cancellation identity is

$$p_a + p_b - p_c = \delta W_0^2 + 2\delta S\Psi W_0 + \delta S^2 P + S^2 E_\Psi.$$

Here $\deg S = 20$, $\deg W_0 = 35$, $\deg \delta = 28$ and $\deg \Psi = 54$. The first three terms have degrees at most $98, 137, 122$ in the rebased frame, with the last bound 124 in the old frame. The final term has degree 148 for Ψ'_{298} and 150 for Ψ'_{300}. Thus the first inner square has degree 296 in the rebased case, while $-4p_ap_b$ has exact degree 298 and cannot cancel. In the old frame the two contributions tie at degree 300, but the inner-square leader has T-degree four and the cross term has T-degree two. Again they cannot cancel. Explicit actual coefficients are

$$[j^{136}w^{46}v^{50}z^6s^6r^{12}c^{40}T^2]\,\mathcal{N}_{298} = 2^{272}3^{182}\beta^{136} > 0,$$
$$[j^{110}w^{50}v^{50}s^4r^{18}c^{64}T^4]\,\mathcal{N}_{300} = 2^{256}3^{160}\beta^{110} > 0.$$

Their exponent sums are 298 and 300. These are attained degrees at the authentic fixed tuple, not generic-port degrees or sampled line tests.

5.7 Two exact degree floors in two distinct fixed frames

Theorem 5.3. *Keep the eleven indeterminates, authentic tuple and all register definitions of the respective scalar frame fixed. Among nonzero constant integer radical weights L and integer polynomial centers Ψ independent of n, the minimum output degree is* 298 *for the rebased frame and* 300 *for the old frame. The displayed centers attain those bounds. Neither statement is an operation-count minimum or a bound for other arithmetic frames.*

Proof. Subscript zero now denotes the complete highest homogeneous part in the actual coordinate ring. Put

$$q_0 = 6\beta j, \quad X_0 = 12q_0w, \quad a_0 = X_0v, \quad F_0 = 12wC - q_0z, \quad R_0 = -q_0^3F_0, \quad \ell_0 = X_0v^2.$$

For the rebased frame, $k_0 = R_0 + h\ell_0$; both terms have degree five. For the old frame $k_0 = R_0$. Set

$$q_* = ra_0^7c^6, \quad s_* = 128sra_0q_0^{15}, \quad p_* = s\ell_0^2k_0^3,$$
$$G_0 = (\text{cone_A}\, vF_0)^2 + (2q_0v^2)^2 + (q_0^3)^2.$$

In the rebased frame, $(\mathcal{B}_6)_0 = -k_0^2G_0$ has degree 16 and $P_0 = -p_*^2G_0$ has degree 54. In the old frame $P_0 = -(q_0^4p_*)^2$ has degree 56. For both, $\delta_0 = q_*$, $S_0 = s_*$, $\deg A_c = 35$, and $\deg(BD_fC_y) = 136$. The complete forms are nonzero at the actual tuple. In particular G_0 is a nonzero sum of three real polynomial squares since $q_0 \neq 0$. The independent top-form checker includes the terms involving h in the rebased k_0 rather than incorrectly reusing the old R_0 alone.

Let $d = \deg \Psi$, with the zero center assigned $d = -\infty$. If $d < 54$, then $\deg Z \leq 73$ and $\deg p_a \leq 174$, while $\deg p_c = 176$ and $\deg p_b = 122$ or 124. Thus the inner sum has degree 176 and the cross term has degree at most 296 or 298, so the output degree is 352. If $d > 54$, then $\deg p_a = 68 + 2d > 176$ dominates, giving $\deg \mathcal{N} = 136 + 4d > 352$; the cross term has smaller degree $190 + 2d$ or $192 + 2d$.

For $d = 54$, let ψ_0 be its nonzero degree-54 form. The rebased cross term has complete degree-298 part

$$(-4p_ap_b)_{298} = (2q_*s_*^2\psi_0p_*)^2G_0.$$

This is a nonzero sum of real polynomial squares. If the inner sum has degree below 149, this proves exact degree 298; if above 149, its square has larger degree. At equality, the degree-298 part is this nonzero sum plus the square of the inner sum's top form and cannot vanish. The old frame has the identical argument at degree 300, replacing G_0 by q_0^8 and the dividing degree 149 by 150. Thus both lower bounds hold. The attained coefficients above supply the matching upper bounds. □

The theorem does not cover changed carriers, polynomial weights, other radical architectures, altered outer-coordinate charts or arbitrary Diophantine encodings. It neither assumes a zero nor decides consistency.

5.8 Measured scalar results and the comparison boundary

Endpoint	Variables	Degree	Operations	Domain
Nearest factorial incoming	11	298	150	$\mathbb{Z}$
Round 19 control	11	300	148	$\mathbb{Z}$
New rebased center	11	298	147	$\mathbb{Z}$
New old-frame center	11	300	145	$\mathbb{Z}$

The degree-298 endpoint uses 29 additions, 24 subtractions and 94 multiplications; the degree-300 endpoint uses 29 additions, 23 subtractions and 93 multiplications. Squares and fixed-scalar multiplications are charged. The degree-298 endpoint improves both degree and cost over the last delivered control, but its novelty against the latest input is a three-gate saving, not a further degree reduction. The degree-300 endpoint is a lower-cost tradeoff. Complete graph and coefficient payloads are supplied; no expanded support or total coefficient height is asserted.

6 Comparison with Round 19 and the latest inputs

6.1 The immutable comparison snapshot

The repository snapshot for this iteration is ProveIt commit

`7a8da414376a80b90ecf799f37ceb26fb9ac674e`,

authored at 22:13:33 UTC on 25 October 2026. The intake ledger binds the Git tree and archive objects. This is newer than both the Round 19 pin `5d22cbf6bf2a344b6abe876e757b3fdc496d4adc` and the `3ecd6a6cf53d47519228328567f45fb7ee9781d9` snapshot used by several newly incoming packages. A comparison against Round 19 alone would omit smaller and lower-degree equations that were already available at the current snapshot.

The seven new packages are Nearest factorial, Digit recovery, Joint norm, Shared factorial error, Factorial-free binomial CRT, Cofinal witnesses and centered carriers, and Lucas corrections and witness elimination. These are separate constructions, not seven names for one source. Joint norm is bound to its delivered Library artifact by its SHA-256 identity; it is not falsely listed as present in the latest Git incoming directory. The report uses the Shared factorial error arithmetic core for its new signed quartic and separately retains the nearest-factorial small-variable branch where explicitly identified. The other packages supply comparison boundaries; their claims are not silently incorporated as premises of our source.

6.2 Direct quartics on unrestricted integers

Every entry in the following table has degree four over $\mathbb{Z}$. Support includes the constant term when nonzero. The first five rows are selected Round 19 endpoints; the remainder are the relevant newly incoming objects.

Endpoint	Variables	Rows	Terms	Height
Round 19 balanced	244	176	5212	3968
Round 19 dimension	243	175	19517	7680
Round 19 sparse	250	182	4770	6709
Round 19 dimension alternative	246	178	21537	6144
Round 19 lower height	247	179	10422	3840
Nearest factorial dimension	242	178	17026	38432
Nearest factorial balanced	243	179	6877	11856
Digit recovery	247	182	4755	8815
Joint norm	245	181	5304	128446
Shared factorial error dimension	240	177	21482	6144
Shared factorial error balanced	241	178	5351	15107
Shared factorial error lower height	241	178	9544	4512
Factorial-free principal	242	178	5018	7968
Factorial-free dimension	241	177	12774	7968
Cofinal witnesses dimension	246	178	12541	12227
Cofinal witnesses sparse	247	179	4352	23316
Cofinal witnesses balanced	247	179	4432	17235
Cofinal witnesses lower height	247	179	5648	6144
Lucas correction principal	248	182	5321	128446
Lucas correction balanced	246	180	5048	195955
Lucas correction dimension	245	179	21845	314701

New Round 20 endpoint	Variables	Rows	Terms	Height
Balanced principal	238	175	5236	3968
Dimension companion	237	174	18569	7680
Power-base companion	239	175	5137	3968

The relevant dimension comparison is therefore not merely the older 243-variable Round 19 equation: the latest Shared factorial error input already reaches 240 variables. The new 237-variable point removes three more coordinates and three rows, and lowers support, but raises height from 6144 to 7680. The new balanced point at 238 variables improves all four measures against that 240-variable input. It does not dominate Round 19's balanced equation because its support is slightly larger. Against the Factorial-free principal, the power-base companion improves variables, rows and height but increases support from 5018 to 5137. The newly incoming sparse Cofinal witness point remains a distinct low-support choice at 4352 terms and a larger coefficient height.

The new arithmetic sphere uses the incoming shared factorial error and signed affine-radix ideas rather than claiming to originate them. Its wider error tolerance, compatibility proof and explicit final reductions are this iteration's additions. The older odd-square restriction is not silently substituted into the new proof. Exact coefficient recollection certifies the entries of the literal source arrays; it does not itself establish every incoming unbounded number-theoretic theorem.

6.3 Height one must be compared on the same source

The Round 19 selected variable/support pairs were

$$(528, 20816), \quad (534, 18543), \quad (580, 12001), \quad (620, 8318), \quad (717, 7176).$$

They all have degree four, unrestricted variables, coefficient height one and constant coefficient one. Section 4 records matched-source contractions separately from graphs compiled from a new arithmetic source. A reduction in the signed source dimension need not reduce the normalized graph dimension or final support: coefficient factorization, monomial incidence, alias multiplicities and graph topology also contribute. A larger support after contraction is reported as a tradeoff rather than suppressed.

6.4 Other domains and higher degrees

The Cofinal witnesses package also gives natural-number quartics at 219/171/8267/130686428, 221/173/5253/65867916 and 221/172/5586/65892620, in variables/rows/support/height order. These are on $\mathbb{N}$, not $\mathbb{Z}$. The Digit recovery package separately reports an unsuccessful three-parameter-calculus experiment at 231/183/6369/83899166 over $\mathbb{N}$; it does not improve its own 229-variable parent. Neither is an unrestricted-integer comparator merely because all coefficients are integers.

The Lucas package supplies monic-elimination tradeoffs on $\mathbb{Z}$ with variable/degree pairs

$$(225, 6),\ (153, 8),\ (144, 12),\ (125, 16),\ (117, 32),\ (111, 64),\ (102, 256),\ (94, 65540).$$

Its 153-variable octic has 188579 collected terms and exact coefficient height 750273658864. These are useful distinct objectives; a smaller variable count at higher degree is not a quartic dimension improvement. The elimination principle is standard and is credited there; we do not relabel these incoming literal circuits as new results of this iteration.

6.5 Small variable circuits and fixed coefficient architectures

In eleven unrestricted variables, Round 19 retained the authentic source34 degree-300, 148-operation point. Nearest factorial already supplies degree 298 at 150 operations by altering the arithmetic budget, as well as fourteen-variable points at degree/cost 82/113 and 90/111, and a thirteen-variable point at 152/126. The Cofinal witnesses package supplies fourteen-variable 88/110 and thirteen-variable 156/117 points. A lower degree at a different dimension is not dominance in every metric.

The strictly positive ten-variable family is another comparison class. Nearest factorial and Cofinal witnesses include degree/cost 382/79; Cofinal witnesses also gives 338/80. Joint norm gives 454/78, and Lucas corrections gives 304/87. Earlier points such as 626/75, 460/77, 328/84 and 288/89 remain separate degree/cost choices. These are incoming compact metrics on their declared positive domain, not new unrestricted-integer results of this report.

Coefficient tuples must remain coherent. The positive-domain radix67 and alternate radix47 encodings, and the signed source34 tuple, have different native meanings. Replacing one port by a smaller coefficient from another tuple does not preserve calibration. Every selected scalar endpoint here retains its specified complete authentic tuple; every independent variable and every fixed integer port is counted distinctly. Collected support and total coefficient height of the scalar equations are not measured unless an actual expansion is supplied.

6.6 The limits of the comparison

The comparison universe consists of named pinned artifacts and prior releases, not every known or possible Diophantine equation. No global minimum in dimension, degree, coefficient height, support or circuit cost is inferred. The direct C222 route is not a K215 simulation or an instantiation of Jones's universal equation; this lineage fact is not a claim to have replaced the inherited logical calibration. Separate reviews are independent AI-assisted checks within this workflow, not external human peer review. Preparing the artifacts does not publish, commit or modify the repository.

7 Reproduction and evidence boundaries

7.1 Portable entry points and complete definitions

From the extracted release root, run

```
python3 -B verify_release.py
```

The integrated verifier checks the release inventory and runs its explicit verification plan in a disposable temporary copy. The option `-output ../receipt.json` saves a fresh receipt outside the sealed release; `-integrity-only` checks inventory rather than mathematical reproduction. Optimized steps set `PYTHONOPTIMIZE=1`, so child Python processes inherit the same mode. This keeps rebuilds and fresh receipts from changing the sealed reference data. The plan and final receipt identify the component commands actually executed. Mathematical checks need Python 3.10 or later and its standard library, not network access, a repository checkout, an external computer algebra system or a proof assistant.

The selected direct equations are `signed/balanced238`, `signed/dimension237` and `signed/power239`. Each contains its entire residual array and its fully collected quartic. `signed/exposed241` and `signed/power_exposed242` are technical source stages, explicitly labeled. The frozen transformation certificates record every source edit, row operation, shear, projection and restoration. A historical coordinate label by itself is not a proof that its semantic value survived an integral translation; the restoration maps supply that interpretation.

The height-one catalog supplies complete final coefficient arrays and ordinary literal polynomial text, not an unevaluated universal-polynomial parameter. Graph certificates specify all aliases, unit paths, internal flows, equality choices, source-row occurrences and accepted deletions. The scalar equations are the complete files `scalar/integer11_degree298_ops147.json` and `scalar/integer11_degree300_ops145.json`, together with their six bound numerical leaves. Registers abbreviate polynomials; only the eleven declared coordinates are independent unknowns.

Rebuild the report with

```
sh report/build_report.sh
```

This needs a standard pdfLaTeX installation. The builder runs three passes in a temporary directory and rejects overfull boxes or unresolved references. The principal residual appendix is generated directly by `report/render_residuals.py`; its binding receipt records source and TeX hashes. PDF metadata and font bytes may vary across TeX installations. The literal coefficient arrays and numerical payloads are the normative mathematical data, not one particular PDF binary.

7.2 Independent finite checks

The checks distinguish source semantics, exact finite algebra and byte identity. Required tests use explicit exceptions and remain active with Python assertions disabled.

- The signed checker imports no producer. It recognizes the exact shared-error source mutation, replays the retained raw projection and translation maps, checks the complete factorial-free power-base and horizon edit, and verifies all source-frame atoms. It authenticates the parent closure, checks unit-pivot restorations, inverse row and coordinate maps, exact final residuals and every collected quartic coefficient. Quadratic restorations require the eliminated coordinate to occur only linearly in the other rows. Malformed coefficients, invalid restorations, noninvertible maps and invalid nonlinear substitutions are rejected.

- The height-one checker independently reconstructs the full energy from divergence squares and equality/unit identities. It checks occurrence coverage, row boundaries, internal trees, alias classes, unit anchoring and all source pullbacks. The new anchored-product equality mode must select an existing unit alias and an alias of its own class. The contraction records must identify a genuine internal flow edge. Universal source-lift identities, complete coefficient height, exact degree, constant and every live coordinate are checked on the actual emitted arrays.

- The scalar checker reconstructs all fixed numerical payloads and checks every live gate in independent free polynomial rings. It proves the exact integral center shears, collects all center-remainder coefficients, and verifies attained actual degree coefficients. A separately

implemented top-form checker binds 23 complete highest homogeneous forms, including all rebased terms involving h. Both alternative operation labels at each of the 292 arithmetic gates are tested: 584 mutated graphs are rejected.

- Deterministic component rebuilds replay frozen finite choices rather than rerunning an optimizer. The normal and optimized-interpreter receipts state the exact checks performed, and the integrated release replay confirms that the portable source closure is sufficient.

The incoming-source inspection receipts are separate from the new endpoint receipts. All seven incoming packages passed their source reconstruction or verification entry points in isolated clean copies, in normal and inherited optimized-Python modes, without altering the original inputs. A separate implementation recollected every coefficient of 60 incoming quartic sums of squares and one 153-variable octic; it also strictly validated 155 residual files containing 205674 literal terms. The scalar numerical reconstruction independently followed 430 machine transitions through 943 tiles and the full convolution and mask recipe. Recollecting an incoming array authenticates its coefficients and measured finite invariants; it does not independently prove all unbounded arithmetic claims of that incoming package.

7.3 The retained logical boundary

The shared direct parent closure is under `signed/parent252`. The complete incoming shared-error proof and the factorial-free proof are retained under `signed/inherited_shared` and `signed/inherited_free`; their historical metrics are not the new endpoint metrics. The newer ordinary delta proofs and frozen transformations identify the exact composition. The C222, ZF and ZFC interfaces remain those named in Section 2.

The height-one outputs inherit precisely their cataloged arithmetic sources. A matched old source is not silently identified with the new 238-variable principal. The scalar branch retains its separate source34 compiler and numerical tuple, the complete rebased incoming proof, and the older source closure. The degree-298 handoff restores the specified B_0-only source; no unproved early cancellation or deleted scale guard is used.

Hashes certify file identity, not mathematical truth. Coefficient replay proves finite identities, not an unbounded Pell or CRT theorem. The new ordinary proofs compose with identified inherited calibrations; neither a Python run nor inclusion of a theorem-name source file rebuilds their full transitive proof-assistant development. No new end-to-end kernel check, encoded object-language ZFC derivation, proof of consistency, or decision of the calibrated equations' solvability is claimed.

8 Further questions

8.1 Can the common norm be weakened again

The new coefficient two is enough because its radius is less than $3/4$, while the true digit and factorial errors are below $1/32$. Replacing it by one would lose this direct uniqueness argument: $1 + 1/32 > 1$. Can retained divisibilities or lattice restrictions exclude the troublesome boundary cases without adding another witness? A useful answer must cover signed and unit radices before positivity is recovered; a small numerical search or a proof assuming the desired signs would not settle that question.

8.2 Can contraction and alias choices be optimized together

An internal-edge contraction changes the cross terms and may make a previously rejected alias identification or square polarization safe. Conversely, an alias choice can block a useful contraction. Can one certify a smaller graph by jointly selecting internal tree edges, source-row orientations and same-logical aliases, while checking every coefficient? An optimum proved within a finite

declared graph family would be a useful result, but it would not be a global minimum among height-one equations.

8.3 Can factorial-free arithmetic improve the same frontier

The incoming factorial-free construction removes factorial authenticity by separate prime-power and label-product arguments. Its source is not a coordinate chart of the shared-error source. A meaningful composition must verify both the reverse CRT construction at all primes and the source-sign bootstrap, then measure the new literal incidence structure. Can it lower dimension without losing the sparse or small-height advantages of the present signed endpoint?

8.4 What changes can lower the small scalar degrees

The retained degree-300 theorem applies only to its fixed arithmetic frame and constant-weight polynomial-center family. A changed budget can fall outside its hypotheses; a coordinate shear can instead lower circuit cost while preserving degree. Further progress should distinguish these mechanisms explicitly. Any proposed carrier or scale change must retain an unbounded witness proof, especially for sign branches and factor cancellation. The rejected early-scale guard deletion is not a usable premise.

8.5 Transporting the higher degree projections

The incoming Lucas package reduces its own 248-variable source by acyclic monic elimination, producing an explicit octic and higher-degree tradeoffs. Can its valid pivot patterns be transported to the new 237- or 238-variable sources, after reconciling their changed coordinates and rows? Such a construction would need new literal restoration certificates, exact degree and live-variable proofs, and measured coefficient bounds or full expansions. No such higher-degree endpoint is claimed in this round; a copied pivot list without those checks would not constitute the required transport.

8.6 A kernel checked bridge from the literal arrays

A focused formalization target is the integral restoration of an actual emitted residual array, followed by the graph-energy zero-set theorem and the existing arithmetic calibration. A transitive proof-assistant build binding the exact array to finite syntactic refutation would strengthen the present ordinary proof and independent finite checks. Included theorem names, checksums and successful Python execution do not already provide that stronger artifact.

A Every residual of the balanced principal equation

The following is the complete finite definition of the principal polynomial. Its independent coordinates are $x_0, \ldots, x_{237} \in \mathbb{Z}$, and

$$P(x_0, \ldots, x_{237}) = \sum_{i=0}^{174} R_i(x)^2.$$

Indices are zero based, exactly as in the literal JSON array. No register or additional quantified variable is implicit in this appendix. The adjacent collected coefficient file specifies the same polynomial.

$$\begin{aligned}
R_0 &= -x_3^2 + x_4 \\
R_1 &= x_0 - x_0x_2 - x_2 + x_9 \\
R_2 &= -1 + x_1 - x_2x_7 - x_3 \\
R_3 &= -1 + x_1 - x_8x_9 \\
R_4 &= 1 + x_4 - x_5x_{20} + x_{18} \\
R_5 &= -1 + x_{10} - x_{10}^2 - 2x_{10}x_{11} - x_{11} - x_{11}^2 \\
&\quad + x_{21} \\
R_6 &= -5 - 20x_{10} - 16x_{10}^2 - 16x_{10}x_{11} - 16x_{10}x_{21} - 10x_{11} \\
&\quad - 4x_{11}^2 - 8x_{11}x_{21} - 10x_{21} - 4x_{21}^2 + x_{22} \\
R_7 &= -2 - x_{10} - x_{10}^2 - 4x_{10}x_{22} - 4x_{22} - 4x_{22}^2 \\
&\quad + x_{23} \\
R_8 &= 1 + 2x_{23} - 4x_{23}^2 + x_{24} \\
R_9 &= -2 - x_{11}^2 - 2x_{11}x_{12} - x_{12} - x_{12}^2 + x_{25} \\
R_{10} &= -9 + 7x_{10} - x_{10}^2 - 4x_{10}x_{11} + 4x_{10}x_{12} - 4x_{10}x_{25} \\
&\quad + 12x_{11} - 4x_{11}^2 + 8x_{11}x_{12} - 8x_{11}x_{25} - 15x_{12} - 4x_{12}^2 \\
&\quad + 8x_{12}x_{25} + 12x_{25} - 4x_{25}^2 + x_{26} - x_{32} \\
R_{11} &= -1 + x_{10} - x_{10}^2 - 2x_{10}x_{12} - x_{11} - x_{12}^2 \\
&\quad + x_{27} \\
R_{12} &= 11 + 6x_{10} - 40x_{10}^2 + 6x_{10}x_{11} - 32x_{10}x_{12} - 32x_{10}x_{21} \\
&\quad + 4x_{10}x_{22} - 16x_{10}x_{27} + 8x_{11}^2 - 8x_{11}x_{12} - 4x_{11}x_{27} + 15x_{12} \\
&\quad - 16x_{12}x_{21} - 9x_{21} - 4x_{21}^2 - 8x_{21}x_{27} + 4x_{22} + 4x_{22}^2 \\
&\quad - x_{23} - x_{26} + 8x_{27} + x_{28} + 2x_{32} - x_{33} \\
R_{13} &= -36x_{12}^2 + 24x_{12}x_{28} - 4x_{28}^2 + x_{29} \\
R_{14} &= -6 - 8x_{10} - 4x_{10}^2 - 8x_{10}x_{11} - 8x_{10}x_{12} + 8x_{10}x_{25} \\
&\quad + 8x_{10}x_{27} - 6x_{11} - 4x_{11}^2 - 8x_{11}x_{12} + 8x_{11}x_{25} + 8x_{11}x_{27} \\
&\quad - 5x_{12} - 4x_{12}^2 + 8x_{12}x_{25} + 8x_{12}x_{27} - x_{21} + 8x_{25} \\
&\quad - 4x_{25}^2 - 8x_{25}x_{27} + 6x_{27} - 4x_{27}^2 + x_{30} \\
R_{15} &= 1 - 4x_{10} + 2x_{11} - 4x_{11}^2 - 8x_{11}x_{12} - 8x_{11}x_{30} \\
&\quad + 4x_{12} - 4x_{12}^2 - 8x_{12}x_{30} - 2x_{21} + 4x_{30} - 4x_{30}^2 \\
&\quad + x_{31} \\
R_{16} &= 29 + 8x_{10} + 10x_{10}^2 + 10x_{10}x_{11} + 4x_{10}x_{12} - 20x_{10}x_{25} \\
&\quad - 16x_{10}x_{27} - 21x_{11} - 13x_{11}^2 - 32x_{11}x_{12} - 40x_{11}x_{25} - 16x_{11}x_{27} \\
&\quad - 11x_{12} - 16x_{12}^2 - 40x_{12}x_{25} - 16x_{12}x_{27} - x_{21} - 40x_{25} \\
&\quad + 4x_{25}^2 + 16x_{25}x_{27} - 12x_{27} + 8x_{27}^2 - 2x_{30} + x_{32} \\
R_{17} &= 6 - 24x_{10} - 8x_{10}^2 + 8x_{10}x_{11} - 16x_{10}x_{12} - 16x_{10}x_{27} \\
&\quad + 13x_{11} - x_{11}^2 + 8x_{11}x_{12} + 4x_{11}x_{27} - 15x_{12} - 8x_{12}^2 \\
&\quad - 16x_{12}x_{27} - 16x_{27} - 4x_{27}^2 - x_{32} + x_{33} \\
R_{18} &= 2 - 28x_{12}^2 - 24x_{12}x_{28} - 32x_{12}x_{33} + 4x_{28}^2 - x_{29} \\
&\quad - 4x_{33}^2 + x_{34} \\
R_{19} &= -3 + 5x_{10} - 4x_{10}x_{11} + 4x_{10}x_{12} + 4x_{10}x_{22} - 4x_{10}x_{25}
\end{aligned}$$

$$
\begin{aligned}
&- 35x_{11} - 32x_{11}^2 + 24x_{11}x_{12} + 24x_{11}x_{21} + 16x_{11}x_{25} - 8x_{12} \\
&+ 4x_{12}^2 + 8x_{12}x_{25} + 16x_{21} - 4x_{21}^2 - 8x_{21}x_{25} + 4x_{22} \\
&+ 4x_{22}^2 - x_{23} + 18x_{25} - 8x_{25}^2 + x_{26} - x_{27} \\
&- x_{32} + x_{35} \\
R_{20} = &-3 - 6x_{10} + 2x_{11} - 2x_{12} - 2x_{27} - 4x_{35} \\
&- 4x_{35}^2 + x_{36} \\
R_{21} = &16 + 30x_{10} + 14x_{10}^2 - 2x_{10}x_{11} - 24x_{10}x_{12} - 16x_{10}x_{21} \\
&+ 60x_{10}x_{22} - 16x_{10}x_{30} + 19x_{11} - 6x_{11}x_{12} - 24x_{11}x_{21} + 32x_{11}x_{22} \\
&- 12x_{12} + 6x_{12}^2 - 12x_{12}x_{21} + 12x_{12}x_{30} + 3x_{21} - 8x_{21}^2 \\
&+ 32x_{21}x_{22} - 8x_{21}x_{30} + 15x_{22} - 4x_{22}^2 - x_{25} - 2x_{27} \\
&+ 4x_{30}^2 + x_{37} + 8x_{40} - 8x_{40}^2 + 2x_{41} - x_{46} \\
R_{22} = &1 + 2x_{37} - 4x_{37}^2 + x_{38} \\
R_{23} = &-x_{11}^2 - 4x_{11}x_{38} - 4x_{38}^2 + x_{39} \\
R_{24} = &-7 + 9x_{10} - 9x_{10}^2 + 22x_{10}x_{11} + 16x_{10}x_{12} - 32x_{10}x_{21} \\
&+ 16x_{10}x_{27} - 6x_{11} - 12x_{11}^2 + 32x_{11}x_{21} - 4x_{11}x_{27} + 14x_{12} \\
&+ 8x_{12}^2 + 16x_{12}x_{27} - 14x_{21} - 16x_{21}^2 - 4x_{25} + 17x_{27} \\
&+ 4x_{27}^2 + x_{32} - x_{33} + x_{40} + 2x_{75} \\
R_{25} = &-4 - 3x_{10} - 7x_{11} - 4x_{12} - 2x_{25} - x_{27} \\
&+ 4x_{40} - 4x_{40}^2 + x_{41} \\
R_{26} = &-6x_{10} + 2x_{11} + 2x_{12} - 4x_{12}^2 - 8x_{12}x_{41} - 2x_{27} \\
&+ 4x_{41} - 4x_{41}^2 + x_{42} \\
R_{27} = &-x_{12} - x_{12}^2 - 2x_{12}x_{13} - x_{13}^2 + x_{43} \\
R_{28} = &-31 + 2x_{10} - x_{10}^2 - 4x_{10}x_{43} - 22x_{11} + 16x_{11}^2 \\
&- 32x_{11}x_{12} + 32x_{11}x_{25} - 42x_{12} - 20x_{12}^2 - 8x_{12}x_{25} + 24x_{12}x_{27} \\
&+ 12x_{25}^2 - 8x_{25}x_{27} + 22x_{27} - 4x_{27}^2 + 2x_{43} - 4x_{43}^2 \\
&+ x_{44} - 4x_{47} \\
R_{29} = &4x_{10} - 2x_{10}^2 - 8x_{10}x_{43} - 16x_{12}^2 - 16x_{12}x_{44} + 4x_{43} \\
&- 8x_{43}^2 - 4x_{44}^2 + x_{45} \\
R_{30} = &-16x_{10}^2 + 24x_{10}x_{12} - 16x_{10}x_{21} + 16x_{10}x_{30} + x_{12} - 9x_{12}^2 \\
&+ 12x_{12}x_{21} - 12x_{12}x_{30} - 4x_{21}^2 + 8x_{21}x_{30} - 4x_{30}^2 + x_{46} \\
R_{31} = &-1 + 6x_{11} - 4x_{11}^2 + 8x_{11}x_{12} - 8x_{11}x_{25} - 6x_{12} \\
&- 4x_{12}^2 + 8x_{12}x_{25} + 6x_{25} - 4x_{25}^2 + x_{47} \\
R_{32} = &-1 - x_{12} - x_{12}^2 - 4x_{12}x_{47} - 4x_{47}^2 + x_{48} \\
R_{33} = &1 + x_{11} - x_{11}^2 - 4x_{11}x_{48} - 4x_{48}^2 + x_{49} \\
R_{34} = &-1 - 6x_{49} - 4x_{49}^2 + x_{50} \\
R_{35} = &-1 - x_{12} - x_{12}^2 - 4x_{12}x_{50} - 4x_{50}^2 + x_{51} \\
R_{36} = &2 + x_{11} - x_{11}^2 - 4x_{11}x_{51} - 4x_{51}^2 + x_{52} \\
R_{37} = &-2x_{46} - 4x_{46}^2 - 8x_{46}x_{52} - 4x_{52}^2 + x_{53} \\
R_{38} = &-9 + 7x_{10} - x_{10}^2 - 4x_{10}x_{53} + 12x_{53} - 4x_{53}^2 \\
&+ x_{54} \\
R_{39} = &-1 - 6x_{54} - 4x_{54}^2 + x_{55} \\
R_{40} = &-x_{12} - x_{12}^2 - 4x_{12}x_{55} - 4x_{55}^2 + x_{56} \\
R_{41} = &-x_{10} - x_{10}^2 - 4x_{10}x_{56} - 4x_{56} - 4x_{56}^2 + x_{57} \\
R_{42} = &-10 - x_{11} + x_{12} - x_{25} - 14x_{57} - 4x_{57}^2 \\
&+ x_{58} \\
R_{43} = &1 - 2x_{11} + 2x_{12} - 2x_{25} + 4x_{58} - 4x_{58}^2 \\
&+ x_{59} \\
R_{44} = &-10 + x_{11} - x_{11}^2 - 4x_{11}x_{59} - 14x_{52} - 4x_{52}^2 \\
&- 4x_{59}^2 + x_{60} \\
R_{45} = &21 + 28x_{52} + 8x_{52}^2 - 2x_{60} - 4x_{60}^2 + x_{61}
\end{aligned}
$$

$$
\begin{aligned}
R_{46} &= 15x_{11} - 36x_{11}^2 + 12x_{11}x_{12} + 24x_{11}x_{21} - x_{12} - x_{12}^2 \\
&\quad - 4x_{12}x_{21} - 5x_{21} - 4x_{21}^2 + x_{62} \\
R_{47} &= 1 + 6x_{11} + x_{12} - x_{12}^2 - 4x_{12}x_{61} - 2x_{21} \\
&\quad - 4x_{61}^2 + 4x_{62} - 4x_{62}^2 + x_{63} \\
R_{48} &= -2 - 2x_{12} + 2x_{12}^2 + 8x_{12}x_{61} + 8x_{61}^2 - 2x_{63} \\
&\quad - 4x_{63}^2 + x_{64} \\
R_{49} &= -2 - 4x_{64} - 4x_{64}^2 + x_{65} \\
R_{50} &= -x_3x_{15} + x_{66} \\
R_{51} &= x_{14} - x_{15} - 2x_{24}x_{66} - x_{67} \\
R_{52} &= 12x_{12}x_{66} + x_{14} - x_{15} - 4x_{26}x_{66} - 2x_{29}x_{66} + 4x_{32}x_{66} \\
&\quad - x_{68} \\
R_{53} &= x_{14} - x_{15} - 2x_{31}x_{66} - 6x_{66} - x_{69} \\
R_{54} &= -12x_{12}x_{66} + x_{14} - x_{15} - 4x_{32}x_{66} - 2x_{34}x_{66} - x_{70} \\
R_{55} &= x_{14} - x_{15} - 2x_{36}x_{66} + 6x_{66} - x_{71} \\
R_{56} &= x_{14} - x_{15} - 4x_{38}x_{66} - 2x_{39}x_{66} - x_{72} \\
R_{57} &= x_{14} - x_{15} - 2x_{42}x_{66} - x_{73} \\
R_{58} &= x_{14} - x_{15} - 2x_{45}x_{66} - 4x_{66} - x_{74} \\
R_{59} &= -x_{10}x_{11} + x_{10}x_{12} + x_{11} + x_{11}x_{12} - x_{12} - x_{12}^2 \\
&\quad + x_{75} \\
R_{60} &= 1 + 2x_{13} + x_{13}^2 - x_{75}^2 + x_{76} \\
R_{61} &= -x_{76}^2 + x_{77} \\
R_{62} &= -4x_{64}x_{66} - 2x_{65}x_{66} + 2x_{66} + x_{78} \\
R_{63} &= -x_{14}^2 + 2x_{14}x_{15} + 2x_{14}x_{78} - x_{15}^2 - 2x_{15}x_{78} - x_{78}^2 \\
&\quad + x_{79} \\
R_{64} &= -1 - x_3x_{11} + x_{80} \\
R_{65} &= -x_{15}x_{80} + x_{82}x_{83} + x_{84} \\
R_{66} &= -1 + x_3 - x_3x_{10} + x_{81} \\
R_{67} &= -1 + 2x_3 + 6x_3x_{11} - 2x_3x_{21} + x_{82} \\
R_{68} &= -x_{14}x_{81} + x_{83} \\
R_{69} &= -6x_3x_{15} + x_{14}x_{80} + 12x_{15} - 8x_{15}x_{80} - 4x_{15}x_{81} - x_{15}x_{82} \\
&\quad + x_{85} \\
R_{70} &= -x_{14}x_{67} + x_{15}x_{67} + x_{86} \\
R_{71} &= -x_{68}x_{86} + x_{87} \\
R_{72} &= -x_{69}x_{87} + x_{88} \\
R_{73} &= -x_{70}x_{88} + x_{89} \\
R_{74} &= -x_{71}x_{89} + x_{90} \\
R_{75} &= -x_{72}x_{90} + x_{91} \\
R_{76} &= -x_{73}x_{91} + x_{92} \\
R_{77} &= -x_{74}x_{92} + x_{93} \\
R_{78} &= -x_{84}x_{93} + x_{94} \\
R_{79} &= -x_{85}x_{94} + x_{95} \\
R_{80} &= -x_{95}^2 + x_{96} \\
R_{81} &= -x_2 - x_2x_{18} + x_{18} + x_{97} \\
R_{82} &= -x_{16}x_{97} + x_{98} \\
R_{83} &= -2x_1^2 + 2x_1x_{14} + 2x_1x_{15} + 2x_1x_{98} + 2x_1x_{99} - x_{14}^2 \\
&\quad - 2x_{14}x_{98} - x_{15}^2 - 2x_{15}x_{99} + x_{19}x_{101} - x_{77}x_{96} - x_{79}x_{96} \\
&\quad - x_{98}^2 - x_{99}^2 \\
R_{84} &= -x_2x_{17} + x_{17} - x_{17}x_{97} + x_{99} \\
R_{85} &= -5x_{10} + 3x_{11} - 4x_{21} - 2x_{24} - 4x_{26} - 2x_{29} \\
&\quad - 2x_{31} - 2x_{34} - 2x_{36} - 4x_{38} - 2x_{39} - 2x_{42} \\
&\quad - 2x_{45} - 4x_{64} - 2x_{65} + x_{100} \\
R_{86} &= -x_{104} - 2x_{104}x_{235} + x_{106} \\
R_{87} &= -x_5x_{128} + x_{102}x_{226} - x_{105} - x_{106}x_{128}
\end{aligned}
$$

$$
\begin{aligned}
R_{88} &= x_{107}x_{227} - x_{109}x_{131} - x_{110} \\
R_{89} &= x_{111}x_{227} - x_{112}x_{131} - x_{113} \\
R_{90} &= x_{114}x_{227} - x_{115}x_{131} - x_{116} \\
R_{91} &= x_{117}x_{227} - x_{118}x_{131} - x_{119} \\
R_{92} &= x_{120}x_{228} - x_{122}x_{131} - x_{123} \\
R_{93} &= x_{124}x_{228} - x_{125}x_{131} - x_{126} \\
R_{94} &= -x_{19}x_{128} + x_{131} \\
R_{95} &= x_{127} - x_{129}x_{131} - x_{130} \\
R_{96} &= -x_{131}x_{133} + x_{132} - x_{134} \\
R_{97} &= -x_{131}x_{136} + x_{135}x_{226} - x_{137} \\
R_{98} &= -x_{131}x_{139} + x_{138}x_{227} - x_{140} \\
R_{99} &= x_{141} - 2x_{141}x_{143} + x_{144} \\
R_{100} &= -x_{141}x_{144} - x_{144} + x_{145} \\
R_{101} &= 2x_{142} + x_{142}^2 + x_{144}^2 - x_{145}^2 \\
R_{102} &= -x_{142}x_{147} - x_{147} + x_{148} \\
R_{103} &= 2x_{146} + x_{146}^2 + x_{147}^2 - x_{148}^2 \\
R_{104} &= x_4^2 - 2x_4x_{142} - 2x_{142} + x_{149} \\
R_{105} &= -1 - x_4x_{147} + x_6 - x_{146} - x_{147} + x_{148} \\
&\quad + x_{149}x_{150} \\
R_{106} &= 1 - 2x_5 + x_5^2 - 2x_5x_{142} + x_{151} \\
R_{107} &= -x_5x_{196} + x_{19} + x_{151}x_{152} - x_{195} + x_{197} \\
R_{108} &= -1 + 2x_{142}x_{211} - x_{153} + 2x_{211} - x_{211}^2 \\
R_{109} &= 1 + x_{153}x_{154} - x_{195} - x_{196}x_{211} + x_{197} + 2x_{235} \\
R_{110} &= -x_{142}x_{156} - x_{156} + x_{157} \\
R_{111} &= 2x_{155} + x_{155}^2 + x_{156}^2 - x_{157}^2 \\
R_{112} &= 1 - 4x_{142} - 4x_{142}x_{235} + x_{158} + 4x_{235} + 4x_{235}^2 \\
R_{113} &= -1 + x_{102} - x_{155} - 2x_{156} - 2x_{156}x_{235} + x_{157} \\
&\quad + x_{158}x_{159} \\
R_{114} &= -1 + x_{103} - 2x_{142}x_{160} - x_{146} - x_{147} - 2x_{147}x_{235} \\
&\quad + x_{148} + x_{158}x_{160} + x_{160} + 4x_{160}x_{235} \\
R_{115} &= 1 + x_{142} - x_{142}x_{162} - x_{162} + x_{163} \\
R_{116} &= 1 + 2x_{161} + x_{161}^2 - 2x_{162} + x_{162}^2 - x_{163}^2 \\
R_{117} &= 1 + x_{107} + x_{158}x_{164} - x_{161} - 2x_{162} - 2x_{162}x_{235} \\
&\quad + x_{163} + 2x_{235} \\
R_{118} &= -2x_{142} - x_{142}^2 + x_{165} \\
R_{119} &= -1 + x_{108} + x_{142}x_{148} - 2x_{142}x_{166} - x_{146} - 2x_{146}x_{235} \\
&\quad - x_{147}x_{165} - 2x_{148}x_{235} + x_{158}x_{166} + x_{166} + 4x_{166}x_{235} - 2x_{235} \\
R_{120} &= -x_{142}x_{168} - x_{168} + x_{169} \\
R_{121} &= 2x_{167} + x_{167}^2 + x_{168}^2 - x_{169}^2 \\
R_{122} &= -1 + x_{111} + x_{158}x_{170} - x_{167} - 2x_{168} - 2x_{168}x_{235} \\
&\quad + x_{169} \\
R_{123} &= -x_{142}x_{172} - x_{172} + x_{173} \\
R_{124} &= 2x_{171} + x_{171}^2 + x_{172}^2 - x_{173}^2 \\
R_{125} &= -1 + x_{114} + x_{158}x_{174} - x_{171} - 2x_{172} - 2x_{172}x_{235} \\
&\quad + x_{173} \\
R_{126} &= -x_{142}x_{176} - x_{176} + x_{177} \\
R_{127} &= 2x_{175} + x_{175}^2 + x_{176}^2 - x_{177}^2 \\
R_{128} &= -1 + x_{117} + x_{158}x_{178} - x_{175} - 2x_{176} - 2x_{176}x_{235} \\
&\quad + x_{177} \\
R_{129} &= -x_{142}x_{180} - x_{180} + x_{181} \\
R_{130} &= 2x_{179} + x_{179}^2 + x_{180}^2 - x_{181}^2 \\
R_{131} &= -1 + x_{120} + x_{158}x_{182} - x_{179} - 2x_{180} - 2x_{180}x_{235} \\
&\quad + x_{181}
\end{aligned}
$$

$$
\begin{aligned}
R_{132} &= -x_{142}x_{184} - x_{184} + x_{185} \\
R_{133} &= 2x_{183} + x_{183}^2 + x_{184}^2 - x_{185}^2 \\
R_{134} &= -1 + x_{121} - 2x_{142}x_{186} + x_{158}x_{186} - x_{183} - x_{184} \\
&\quad - 2x_{184}x_{235} + x_{185} + x_{186} + 4x_{186}x_{235} \\
R_{135} &= -x_{142}x_{188} - x_{188} + x_{189} \\
R_{136} &= 2x_{187} + x_{187}^2 + x_{188}^2 - x_{189}^2 \\
R_{137} &= -1 + x_{124} + x_{158}x_{190} - x_{187} - 2x_{188} - 2x_{188}x_{235} \\
&\quad + x_{189} \\
R_{138} &= -x_{142}x_{192} - x_{192} + x_{193} \\
R_{139} &= 2x_{191} + x_{191}^2 + x_{192}^2 - x_{193}^2 \\
R_{140} &= -1 + x_{127} + x_{158}x_{194} - x_{191} - 2x_{192} - 2x_{192}x_{235} \\
&\quad + x_{193} \\
R_{141} &= 1 - x_{142}x_{196} - x_{196} + x_{197} \\
R_{142} &= -1 + 2x_{195} + x_{195}^2 + x_{196}^2 - 2x_{197} - x_{197}^2 \\
R_{143} &= x_{128} - 2x_{142}x_{198} + x_{158}x_{198} - x_{195} - x_{196} - 2x_{196}x_{235} \\
&\quad + x_{197} + x_{198} + 4x_{198}x_{235} \\
R_{144} &= -x_{142}x_{200} - x_{200} + x_{201} \\
R_{145} &= 2x_{199} + x_{199}^2 + x_{200}^2 - x_{201}^2 \\
R_{146} &= -1 + x_{132} + x_{158}x_{202} - x_{199} - 2x_{200} - 2x_{200}x_{235} \\
&\quad + x_{201} \\
R_{147} &= -x_{142}x_{204} - x_{204} + x_{205} \\
R_{148} &= 2x_{203} + x_{203}^2 + x_{204}^2 - x_{205}^2 \\
R_{149} &= -1 + x_{135} + x_{158}x_{206} - x_{203} - 2x_{204} - 2x_{204}x_{235} \\
&\quad + x_{205} \\
R_{150} &= -x_{142}x_{208} - x_{208} + x_{209} \\
R_{151} &= 2x_{207} + x_{207}^2 + x_{208}^2 - x_{209}^2 \\
R_{152} &= -1 + x_{138} + x_{158}x_{210} - x_{207} - 2x_{208} - 2x_{208}x_{235} \\
&\quad + x_{209} \\
R_{153} &= -10 - 2x_0 - 3x_0^2 - 2x_0x_4 - x_1^2 - x_2^2 \\
&\quad - 2x_4 - 2x_4^2 - x_5^2 - x_6^2 + 2x_{10} - x_{10}^2 \\
&\quad - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 \\
&\quad - x_{17}^2 - x_{18}^2 - x_{19}^2 - x_{100}^2 - x_{102}^2 - x_{103}^2 \\
&\quad - x_{107}^2 - x_{108}^2 - x_{111}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 \\
&\quad - x_{121}^2 - x_{124}^2 - x_{127}^2 - x_{128}^2 - x_{132}^2 - x_{135}^2 \\
&\quad - x_{138}^2 + x_{141} - x_{211}^2 - 12x_{235} - 8x_{235}^2 \\
R_{154} &= -x_5x_{10} - x_5x_{11} - x_5x_{12} - x_5x_{13} - x_5x_{14} - x_5x_{15} \\
&\quad - x_5x_{16} - x_5x_{17} - x_5x_{18} + x_{211} \\
R_{155} &= -16 + x_0 - x_{212}^2 \\
R_{156} &= -1 + x_2 + x_6 - x_6^2 \\
R_{157} &= -x_{142}x_{214} - x_{214} + x_{222} \\
R_{158} &= -1 + x_{213}^2 + x_{214}^2 - x_{222}^2 \\
R_{159} &= -x_{216} - x_{216}x_{217} + x_{223} \\
R_{160} &= -1 + x_{215}^2 + x_{216}^2 - x_{223}^2 \\
R_{161} &= -x_{208}^2 + x_{224} \\
R_{162} &= x_{214} - 2x_{218}x_{224} - x_{224} \\
R_{163} &= x_{142}x_{213} + x_{219} \\
R_{164} &= -x_{142} - x_{213}x_{219} + x_{217} \\
R_{165} &= -1 - x_{207} - x_{213}x_{220} + x_{215} \\
R_{166} &= -x_{100} - x_{208}x_{221} + x_{216} \\
R_{167} &= 2x_0^2 + 2x_0x_4 - 2x_0x_{147} - 2x_0x_{156} + x_1^2 - 2x_1x_{196} \\
&\quad + x_2^2 - 2x_2x_{184} + x_4^2 - 2x_4x_{156} + x_{10}^2 - 2x_{10}x_{162} \\
&\quad + x_{11}^2 - 2x_{11}x_{168} + x_{12}^2 - 2x_{12}x_{172} + x_{13}^2 - 2x_{13}x_{176}
\end{aligned}
$$

$$
\begin{aligned}
&+ x_{14}^2 - 2x_{14}x_{180} + x_{15}^2 - 2x_{15}x_{188} + x_{16}^2 - 2x_{16}x_{192} \\
&+ x_{17}^2 - 2x_{17}x_{200} + x_{18}^2 - 2x_{18}x_{204} + x_{100}^2 - 2x_{100}x_{208} \\
&- x_{142}x_{225} + x_{146}^2 + x_{147}^2 + x_{155}^2 + x_{156}^2 + x_{161}^2 \\
&+ x_{162}^2 + x_{167}^2 + x_{168}^2 + x_{171}^2 + x_{172}^2 + x_{175}^2 \\
&+ x_{176}^2 + x_{179}^2 + x_{180}^2 + x_{183}^2 + x_{184}^2 + x_{187}^2 \\
&+ x_{188}^2 + x_{191}^2 + x_{192}^2 + x_{195}^2 + x_{196}^2 + x_{199}^2 \\
&+ x_{200}^2 + x_{203}^2 + x_{204}^2 + x_{207}^2 + x_{208}^2 \\
R_{168} &= -x_{103}x_{226} + x_{128} \\
R_{169} &= -x_{108}x_{227} + x_{128} \\
R_{170} &= -x_{121}x_{228} + x_{128} \\
R_{171} &= -2 - x_{100}^2 - x_{147}^2 - x_{156}^2 + 2x_{162} - x_{162}^2 \\
&- x_{168}^2 - x_{172}^2 - x_{176}^2 - x_{180}^2 - x_{184}^2 - x_{188}^2 \\
&- x_{192}^2 - x_{196}^2 - x_{200}^2 - x_{204}^2 + x_{208}^2 - x_{229}^2 \\
&- x_{230}^2 - x_{231} - x_{231}^2 \\
R_{172} &= -2x_{105}^2 - 2x_{110}^2 - 2x_{113}^2 - 2x_{116}^2 - 2x_{119}^2 - 2x_{123}^2 \\
&- 2x_{126}^2 + x_{128}^2 - 2x_{130}^2 - 2x_{134}^2 - 2x_{137}^2 - 2x_{140}^2 \\
&- x_{232}^2 - x_{233}^2 - x_{234} - x_{234}^2 - 2x_{237}^2 \\
R_{173} &= -x_5x_{236} + x_{128} \\
R_{174} &= x_3x_{128} - x_6x_{236} - x_{237}
\end{aligned}
$$